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Rules for the calculation of the Carbon Footprint of industrial Batteries without external storage (CFB-IND)

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Abstract

This report provides the methodological guidelines for calculating the carbon footprint of industrial batteries (CFB-IND), providing the basis for the enforcement of requirements as in Article 7 of Regulation (EU) 2023/1542 (Batteries Regulation), establishing the methodology for calculation and verification of the carbon footprint of batteries. The present report is targeting industrial batteries with exclusively internal storage and with an energy storage capacity >2 kWh.

It is based on the previous JRC draft report that was distributed in June 2024 as a basis for a second stakeholder consultation on the draft CFB-IND methodology and on the draft Annex to the Delegated Act on the carbon footprint methodology for electric vehicle batteries (draft DA-EV) as published on "Have your Say" on April 30th 2024. In an attempt to harmonize the CFB methodologies as far as possible, maximum agreement with the draft DA-EV was aspired, keeping the wording as close as possible over large passages. Stakeholder comments provided for the draft report on the CFB-IND rules, those submitted on the draft DA-EV, and feedback received in the subsequent interaction with stakeholder have been evaluated and incorporated when deemed appropriate.

The main differences to the draft DA-EV are given in the definition of the system boundaries and the functional unit, with (i) a distinction between custom-made and off-the shelf batteries that does not exist for electric vehicle batteries, and (ii) a functional unit linked to the performance parameters to be declared under Article 10 of Regulation (EU) 2023/1542, specifically the declared lifetime in cycles for batteries providing repetitive energy supply (REP) and the declared lifetime in years for on-demand batteries (OND). Several new default recycling models have further been introduced due to the broader scope of the CFB-IND, covering also relevant industrial battery chemistries such as lead-acid, nickel-cadmium and sodium-sulphur batteries.

As such, the present report provides recommendations for a consistent approach for determining the CFB-IND. It will serve as a technical basis for the further policy decisions on the final method for determining the CFB-IND.

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We would like to thank our colleagues Conrad Szczuka and Matthias Bruchhausen from JRC Petten for the help and continuous fruitful exchange on battery durability and lifetime estimation, without these we would not have been able to solve many of the tricky aspects related with the functional unit. We also thank our recycling team, specifically Silvia Bobba, Martina Orefice and Fabio Manni for their support in understanding the details of battery recycling processes, and Cynthia Latunussa for all the modelling efforts that helped us to better understand the impacts of battery recycling and secondary materials on the carbon footprint. Thanks also to Enrica Leccisi for helping to solve the challenges related with datasets and methodological alignment.

We further acknowledge the input provided by numerous stakeholders that helped to improve the methodology and to spot aspects that were not considered in previous versions. Despite the huge number of comments received, we analysed all and tried to address them as far as possible and reasonable. Specific thanks in this regard to those stakeholders who provided primary data and who supported in modelling, thus closing data gaps and improving the quality of the report.

Finally, thanks also to Lisa Della Pietra and Ewout Deurwaarder from the Directorate General for Internal Market, Industry, Entrepreneurship and SMEs (DG-GROW) and Rana Pant from the Directorate-General for Environment (DG-ENV) for support and input on policy developments and the Regulation text, and all other colleagues from the EC that are not mentioned explicitly but contributed with valuable input and expertise.

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Executive summary

This report provides methodological guidelines for calculating the Carbon Footprint of Industrial Batteries (CFB-IND), in line with the objectives of Article 7 of Regulation (EU) 2023/1542 ('EU Batteries Regulation') (European Union, 2023). It is based on the Environmental Footprint (EF) method developed by the European Commission (Commission Recommendation (EU) 2021/2279) (European Commission, 2021), and the draft Delegated Act that defines the methodology for calculating the carbon footprint of electric vehicle batteries (draft DA-EV), as published in April 2024 for public consultation on "Have your Say"[1] (European Commission, 2024). At the moment of writing the present report, the work on the draft DA-EV is still ongoing and the final version is expected in 2025. Due to the different nature of industrial batteries and their very wide field of applications, some substantial modifications have been made in comparison to the draft DA-EV. This holds true above all for the functional unit, the system boundaries and the End of Life (EoL) modelling, since the category of industrial batteries comprises virtually any battery with an energy capacity above 2 kWh that does not fall under any of the remaining battery categories defined in Regulation (EU) 2023/1542, namely electric vehicle batteries, Light Means of Transport (LMT) batteries, and Starter, Lighting, or Ignition (SLI) batteries. Batteries with external storage, while considered industrial batteries under Regulation (EU) 2023/1542, are not covered by the present report since a separate delegated act will be drawn up for these at a later stage. The application fields of industrial batteries with internal storage range from stationary backup systems over large containerised energy storage, home storage systems, until airborne and maritime batteries used for either propulsion or other services. A major challenge was therefore to define methodological guidelines that are suitable and appropriate for all these diverse batteries and corresponding applications.

The report is comprised of seven sections, which have been aligned as far as possible with the structure of the draft DA-EV, thus facilitating the comparison. The major differences in comparison with the draft DA-EV are summarized in the following.

Scope:

Evidently, the scope of the present document differs from that of the draft DA-EV. Industrial batteries comprise a larger number of possible cell chemistries and a wide field of different applications, from stationary to airborne mobile batteries, and from backup systems to intensively cycled energy applications. While it turned out that for the calculation rules there is no need for distinguishing all these different battery chemistries and applications, there is a clear split between off-the shelf (OtS) batteries and custom-made (CM) batteries because the way how these systems are designed and developed is fundamentally different. For CM batteries, the final application and its load profiles are well known and specified in detail when placing the battery on the market, while OtS batteries are designed for an assumed average use-case, with the actual load profile mostly unknown *a priori*. Also, a broader set of cell chemistries was considered when drafting the present rules, including (beyond the lithium-ion and nickel-metal hydride batteries previously considered for electric vehicle batteries) lead-acid, nickel cadmium, sodium-ion and high-temperature sodium-sulphur batteries.

[1] https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13877-Batteries-for-electric-vehicles-carbon-footprint-methodology_en

Calculation Rules:

For the calculation rules, an additional distinction is made between on-demand (OND) batteries and repetitive energy supply (REP) batteries, as also reflected in the Regulation text (Annex II) where a different functional unit is defined for backup batteries. Correspondingly, the declaration of the CFB and of its underlying functional unit i.e., the unit of comparison for the CFB, differs between REP and OND batteries. The functional unit is among the most debated aspects of the CFB rules and has been discussed extensively with stakeholders, suggesting and evaluating a variety of different options. The warranty-based approach, as followed in the draft DA-EV defining the methodology for the carbon footprint of electric vehicle batteries, was found not to be applicable to industrial batteries, especially in the case of CM batteries. Hence, the functional unit for industrial batteries is based, in this report, on the lifetime parameters as to be declared under Article 10 of Regulation (EU) 2023/1542, particularly the expected lifetime in cycles for REP batteries and the expected lifetime in years for OND batteries. However, since no standards are yet available that define how the Article 10 parameters are to be determined, own definitions and requirements are provided to narrow the potential divergence of lifetime declarations. These definitions and requirements are aligned as far as possible with current standardisation activities and upcoming implementing legislation on durability. The key parameters of the expected lifetime declaration must be disclosed publicly in the CFB supporting study with the aim of creating transparency and discourage unrealistic declarations.

System Boundaries and cut-off rules:

While the system boundaries are defined at a higher level by the Regulation text and are therefore identical between electric vehicle (EV) batteries and industrial (IND) batteries, in practice their definition requires more detail for IND batteries due to their heterogeneous layout. IND batteries are not necessarily contained in a single housing and may require additional or external components or replacement parts that have to be considered. Again, a distinction is made between CM and OtS batteries, with the former relying on a clear customer requirement and a specification to fulfil this requirement including all relevant components. Here, verifiability is ensured by disclosing the underlying specifications to the notified body, and by providing the key parameters in the public version of the CFB supporting study.

Regarding the life cycle stages and cut-off rules, little discrepancy is given compared to the rules for the Carbon Footprint of Electric Vehicle Batteries (CFB-EV), since the underlying philosophy is the same. Only some additional materials and components are considered due to the broader scope of CFB-IND and the additional cell chemistries such as lead-acid (PbA), nickel-cadmium (NiCd) or sodium-sulphur (Na-S) high -temperature batteries.

Data collection requirements and quality requirements:

As for the system boundaries, the data collection is almost identical to the procedure defined by the CFB-EV, except for some additional materials defined as most relevant (due to the larger variety of cell chemistries). As for EV batteries, mandatory company-specific processes (all processes that belong to the manufacturing life cycle stage) are distinguished from non-mandatory company-specific processes, with the latter again separated into most relevant and non-most relevant processes. Data collection requirements vary depending on the type of process (see next paragraph).

Inventory data - Modelling requirements:

In terms of modelling requirements, the main difference to the draft DA-EV is given in the sections on recycling modelling and on the circular footprint formula, again due to the additional cell

chemistries that are considered. Specific default recycling models have been developed based on the input from several stakeholders, literature reviews and exchange with internal experts. These recycling models are provided as tables in Section 6.3 and represent current average industry practice. As in the draft DA-EV defining the CFB methodology for EV batteries, they therefore represent a conservative model, giving manufacturers the option to declare own, better performing recycling processes. The return rate, indicating the fraction of batteries that are recovered for ordinary recycling in Europe at the end of their life and that are treated with the corresponding recycling processes under EU legislation is set to 95% across all industrial batteries. The other parts of the circular footprint formula remain unchanged. However, an additional section has been added on the modelling of secondary materials. While this does not constitute diverging requirements, this aspect is not described explicitly in the draft DA-EV and additional clarification on how to model such cases were deemed advantageous, as also reflected by stakeholder feedback.

Verification:

Regarding the verification process and requirements, no additional or diverging requirements were added in comparison to draft DA-EV defining the CFB methodology for EV batteries. However, there are important differences in terms of the content of the CFB supporting study and of its corresponding public version that both have to be provided by the manufacturer. This is due to the different functional unit and the need to make the key parameters used for the lifetime declaration (such as lifetime in cycles, useable energy, reference tests) publicly accessible and therefore as transparent as possible.

The present report is the result of extensive exchange with numerous stakeholders over the time period 2023 - 2024, including two online workshops held in December 2023 and in July 2024 and further dedicated exchanges with individual stakeholders. All feedback has been analysed and used, when deemed appropriate, for the proposed approach for accounting the CFB-IND. As such, the present report represents the outcome of a consensus building process on CFB based on different available sources and various positions from different stakeholders.

1 Introduction

The EU Batteries Regulation (Regulation (EU) 2023/1542) published the 28th of July 2023 (European Union, 2023) sets rules on sustainability, performance, safety, collection, recycling, and second life of batteries as well as on information requirements about batteries. Among these, there is the carbon footprint of the batteries (CFB) to be determined and communicated as specified in Article 7 and in the corresponding Annex II of Regulation (EU) 2023/1542. The CFB quantifies the total amount of greenhouse gases emitted along the life-cycle of the battery, in terms of kg of CO_2 equivalent per functional unit. The functional unit is defined by the Annex II of the Regulation as one kWh (kilowatt-hour) of the total energy provided by the battery system over the battery's service life, measured in kWh. For back-up batteries that have as their primary function to ensure continuity of a power source, the functional unit is defined as the ability to provide one kW·min (kilowatt-minute) of backup power capability at any moment over the lifetime of the battery.

According to Article 7, the CFB shall be declared for "rechargeable industrial batteries with a capacity above 2 kWh, light means of transport (LMT) batteries and electric vehicle (EV) batteries placed on the Union market". CFB declarations will then be used to define the CFB performance classes and CFB thresholds that batteries entering the European market should comply with. Ultimately, Regulation (EU) 2023/1542 will ensure that the expected massive deployment of batteries will be associated with minimum overall carbon emissions. The CFB shall be accompanied by a public version and a non-public version of the CFB supporting study.

The Joint Research Centre (JRC) of the European Commission has been in charge of providing technical support to the development of the secondary legislation on the CFB (in line with the requirements of the Article 7 and Annex II of Regulation (EU) 2023/1542). The present document lays down the draft methodology and the technical and scientific base for determining the CFB of "rechargeable industrial batteries with a capacity above 2 kWh, excluding those with exclusively external storage" (hereafter abbreviated as CFB-IND). As such, it is conceived as a technical basis for the further policy developments on the Delegated Act(s) that will ultimately define the official methodology to be followed for the CFB declaration under Article 7 of Regulation (EU) 2023/1542.

It is acknowledged that some components of the CFB method are common between all types of batteries in scope. For such reason the present report is based on the Annex to the draft Delegated Act that lays down the methodology for calculating and verifying the life cycle carbon footprint of electric vehicle batteries, as published for public consultation on 'Have your Say' in April 2024 (draft DA-EV)[2] (European Commission, 2024). Since June 2024, a draft of the Carbon Footprint rules for Industrial Batteries (CFB-IND rules) has been also published, as draft for stakeholders' comments, on the European Platform on LCA[3] (European Commission: Joint Research Centre, European Platform on LCA (EPLCA), 2024).

Note: The text of carbon footprint rules has been kept as concise as possible. All sections specified as "Note" have to be considered as explicative to the text and not be intended as part of the rules for Carbon Footprint of Batteries (CFB).

[2] https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/13877-Batteries-for-electric-vehicles-carbon-footprint-methodology_en

[3] https://eplca.jrc.ec.europa.eu/EU_BatteryRegulation_Art7.html

Unless otherwise specified in this document, the EF method as in the EC Recommendation 2021/2279 (European Commission, 2021) shall be considered as general guidelines about how to determine the carbon footprint.

1.1 Terminology

This report uses the following terminology to indicate the requirements, as:

- The term “shall” is used to indicate what is required in order to calculate the CFB.
- The term “may” is used to indicate an option that is permissible. Whenever options are available, the CFB supporting study shall include adequate argumentation and evidence to justify the chosen option.

2 Scope

This document provides the draft methodological guidelines and the technical and scientific base for calculating the carbon footprint of rechargeable industrial batteries with a capacity above 2 kWh, excluding those with exclusively external storage (CFB-IND) within the scope of the Article 7 of Regulation (EU) 2023/1542, as it will be ultimately defined in the corresponding Delegated Act setting the official methodology for determining the CFB-IND. The CFB-IND shall be declared for any industrial battery that falls within the scope of Article 7, independently of its cell chemistry and of the specific application. The scope of the CFB-IND excludes batteries that have been subject to preparation for re-use, preparation for repurposing, repurposing, or remanufacturing, and that had already been placed on the market or put into service before undergoing such operations (according to the Article 7 (point 5) of Regulation (EU) 2023/1542).

Note: Although the rules defined in this document are generic and applicable to any rechargeable industrial battery with internal storage, independently of its chemistry, the following battery chemistries have been analysed for the testing and development of the present CFB-IND rules (especially related to Sections 5.2 and 6.3), being currently considered the most relevant ones:

- *Lead-Acid (flooded, valve-regulated / sealed, tubular and all other forms - PbA);*
- *Lithium-based, including all current lithium-ion battery (LIB) chemistries and solid-state LIB;*
- *Sodium High Temperature (Na-S, Na-NiCl);*
- *Sodium-Ion (SIB), including aqueous SIB and hybrid sodium/lithium-ion batteries*
- *Nickel-Cadmium (NiCd), Nickel-Zinc (NiZn) and Nickel-Metal Hydride (NiMH).*

Other emerging chemistries like Magnesium-Sulfur (Mg-S), Aluminium, etc. could potentially become relevant in future, especially once achieving market maturity. The same applies to emerging anode / cathode materials for any of the listed battery chemistries and to other battery technologies. If deemed necessary, the CFB rules may be updated correspondingly.

Given the broad field of application for industrial batteries, the present report distinguishes between batteries that are **custom-made (CM)** and those that are commercialised "**off the shelf" (OtS)**:

a. **CM batteries** are designed and developed for a specific application in a given location according to the customer requirements described in their technical purchasing specification (TPS). For being considered CM, the battery manufacturer shall provide the corresponding evidence in the CFB supporting study, including the final customer issued TPS, and the auditable technical specification (ATS) as provided by the manufacturer to the customer together with the delivered battery, such as an auditable battery sizing report. The TPS shall reflect the exact configuration and requirements of the application as agreed on and referred to in the contractual purchase (or equivalent) agreement, and the ATS shall reflect the exact configuration and parameters of the battery as delivered to the customer. It shall be referred to explicitly in the underlying contractual agreement.

b. **OtS batteries** are produced in batches or series of uniform configuration and sold to the client without being certain of the final application or use-profile beforehand. Typical examples would be home storage systems or large-scale container systems. OtS batteries are designed for a typical (reference) use profile as defined by the manufacturer according to the foreseen main application.

3 Calculation Rules

Two fundamentally different services can be provided by rechargeable industrial batteries, independently of whether they are CM or OtS-type batteries:

1. The **provision of electricity** that was previously charged into the battery. While different applications have very different use profiles, they all require the battery to regularly discharge its energy content, leading to a more or less frequent cycling behaviour. Systems providing this type of service are considered to provide **repetitive energy supply (REP)**.

2. The **provision of back-up** or on-demand **services**, maintaining the battery mostly in charged state and discharging only occasionally, e.g. in case of power failure or for system start-up. In the extreme case, the grid failure or power outage never happens, and the battery is never cycled but just kept as safety reserve. Systems providing this type of service are referred to as **on-demand (OND) batteries**. For being considered OND, a battery shall be explicitly declared as such in its technical documentation and provide less than 40 full equivalent cycles (FEqC) per year in the application.

Note: The classification is based on the IEC 63369 under development.

3.1 Life Cycle Impact assessment and Reference Flow

The CFB-IND shall be calculated as the amount of CO_2-equivalents emitted during the life cycle stages of the battery that are within the system boundary, expressed in kilogram ('kg') CO_2-equivalent, divided by:

(i) for REP batteries: the total amount of energy provided by the battery over the battery's service life;

(ii) for OND batteries: the total amount of backup service provided by the battery over its service life, determined in accordance with Section 3.2.

The carbon footprint shall be reported in kg CO_2-equivalents/ kilowatt-hour ('kWh') (for REP batteries), or in kg CO_2-equivalents /kW·min·y (for OND batteries), with a resolution of two significant digits.

The amount of kg CO_2-equivalents emitted shall be determined by identifying the emissions of greenhouse gases related to the materials and energy used and, where relevant, produced in the life cycle stages of the battery that are within the system boundary, in accordance with Sections 4.1 and 4.2. The amount of kg CO_2-equivalents shall be calculated from these emissions of greenhouse gases by applying the Environmental Footprint ('EF') 3.1 impact assessment method available in the Life Cycle Data Network (LCDN) for the impact category "climate change".

All quantitative input and output data collected to quantify the carbon footprint shall be calculated in relation to the reference flow. The reference flow shall be calculated as the total mass of battery divided by the total amount of energy provided by the battery over the battery's service life, and thus measured in kg/kWh (REP), or as the total mass of battery divided by the total backup service provided by the battery over its service life, and thus measured in kg/kW·min·y (OND).

If, over the course of time, due to changes in the bill of materials (i.e., the list of all material inputs needed to manufacture the product in scope of the study), changes in the origin of the materials, changes in processes, changes related to the use of electricity and other auxiliaries, or any other

changes, the CFB increases by more than 10% compared to the carbon footprint calculated, this shall be considered a change to the battery's technical characteristics relevant for the requirements of Regulation (EU) 2023/1542 and thus for the new battery model a new carbon footprint shall be calculated and a new carbon footprint declaration shall be drawn up.

3.2 Functional unit

3.2.1 Functional unit for repetitive energy supply (REP) industrial batteries

For energy-providing (REP) batteries, the **functional unit** is defined (according to Annex II of Regulation (EU) 2023/1542) as "*one kWh (kilowatt-hour) of the **total amount of energy** delivered by the battery over its service life, measured in kWh".*

3.2.1.1 Off the Shelf (OtS)-type REP industrial batteries

For OtS-REP batteries, the **total amount of energy** provided by the battery over the battery's service life ('E_{total}'), expressed in kWh, shall be calculated as follows:

$$E_{total} = initial\ useable\ energy\ capacity \cdot cycles\ provided$$

Where:

- ***Initial useable energy capacity*** is the energy available to the user when discharging a fully charged, new battery (at beginning of its service life) within the charge/discharge limits set by the battery management system under standard operating conditions. "Fully charged" refers to charging until the charge level set by the battery management system under standard operating conditions.
- ***Cycles provided*** is the battery expected lifetime expressed in full equivalent cycles (FEqC), as declared according to Article 10 / Annex IV. The values shall correspond with those reported in the technical specification delivered with the battery.

  If the battery expected lifetime is not expressed in FEqC, but in **full cycles** within the charge/discharge limits set by the battery management system under standard operating conditions, the number of FEqC may be estimated by multiplying the number of full cycles with the mean value between the initial state of health and the state of health at the end of the battery's service life (linear interpolation).

  If the battery expected lifetime is expressed only in **partial cycles**, the number of FEqC may be estimated by multiplying the number of partial cycles with the ratio between the average energy delivered per partial cycle over service life and the initial useable energy capacity.

  The exact test procedure (including charge-discharge profiles over time, C-rates, temperature), the standard operating conditions and, if applicable, the underlying standard used for determining the battery expected lifetime shall be provided in the public version of the CFB supporting study. The threshold for determining the end of service life of the battery shall be equal or above 70% of the battery's initial useable energy capacity. "End of service life" refers to the end of the battery's first service life, i.e. within the primary application for which it was originally designed; any second life is not considered.

  If the value of '*cycles provided*' is higher than the lifetime in years (as declared according to Article 10 / Annex IV) multiplied with 1000 cycles/year, then a figure equal to 'lifetime in

years * 1000 cycles/y' shall be used for '*cycles provided*' in the calculation of the total amount of energy.

Note: This impedes too extreme discrepancies between service life in cycles and service life in years. 1000 FEqC/y is considered to be at the upper end of typical use profiles.

The total amount of energy, the initial useable energy capacity, the lifetime in cycles and in years, and the threshold for determining the end of service life shall be provided in the public version of the CFB supporting study.

Note: In case there was a need for a default value for the cycles provided (e.g., to cover the possibility that the corresponding documentation is not provided or that the above-mentioned requirements are not met), a default value of 400 FEqC/y is recommended for 'cycles provided'.

Note: The value of 400 FEqC/y is adopted from the Product Environmental Footprint Category Rules (PEFCR) for rechargeable batteries (RECHARGE, 2024) *and in line with the JRC Report on Performance and Durability Requirements* (European Commission: Joint Research Centre, Szczuka, C., Sletbjerg, P. and Bruchhausen, M.,, 2024).

*Note: **FEqC** is the number of times the battery would theoretically be able to discharge its initial useable energy capacity. The FEqC multiplied with the initial useable energy capacity gives the total energy provided by the battery over lifetime.*

*Note: **Full cycles** is the number of times the battery can fully discharge within the charge/discharge limits set by the battery management system under standard operating conditions. The actual energy delivered in each full cycle decreases over lifetime. For a new battery (at beginning of its service life), one full cycle is equivalent to one FEqC. The full cycles multiplied with the initial useable energy capacity and the average value between 100% and the state of health (or remaining useable energy capacity in %) at EoL gives the total energy provided by the battery over lifetime.*

*Note: **Partial cycles** is the number of times the battery can discharge with a given discharge profile, but without fully discharging within the charge/discharge limits set by the battery management system.*

3.2.1.2 Custom-made (CM)-type REP industrial batteries

For CM-REP batteries, the **total amount of energy** provided by the battery over the battery's service life ('E_{total}'), expressed in kWh, shall be taken from the final technical purchasing specification (TPS) that is underlying the corresponding contractual agreement. The corresponding auditable technical specification (ATS) delivered with the battery shall explicitly refer to the final TPS and list all replacement parts that are needed so that the battery as delivered can fulfil the application requirements as provided in the TPS. Both the TPS and the ATS including the underlying application load profile shall be fully disclosed in the CFB supporting study.

The total amount of energy, the the initial useable energy capacity, the lifetime in cycles and in years shall be provided, in both the TPS and the ATS. Their values shall correspond to the values declared according to Article 10 / Annex IV of Regulation (EU) 2023/1542 and be disclosed in the public version of the CFB supporting study. In case of discrepancy between the value provided for any of these parameters in the TPS and the ATS, the lower value shall be used.

Note: In case of CM batteries, the values specified in the TPS (the application requirements) may differ from those specified in the ATS (the battery specification) such as in case a battery with very

high cycles is installed in an application that does not require that many cycles. In this case, the lower value shall be used since it reflects the actual total amount of energy provided by the battery.

In case the corresponding documentation is not provided, or it does not fulfil the above-mentioned requirements, the battery shall be treated like an OtS battery.

3.2.2 Functional unit for on-demand (OND) or backup industrial batteries

On-demand batteries (such as uninterruptible power supply - UPS) are only occasionally or even never cycled, their principal function being the ability to provide power for short times e.g., in case of power outage.

The **functional unit** for the CFB of OND batteries is defined as (Regulation (EU) 2023/1542):

"Provision of 1 kW·min of backup power capability over the whole service life. Provision of backup power capability refers to the ability to discharge at the given power and for the given backup time without interruption at continuous discharge above minimum voltage at any moment over the lifetime of the battery".

Correspondingly, the reference flow is measured in kg of battery per kW·min·y of **total backup service** provision.

3.2.2.1 Off the Shelf (OtS)-type OND industrial batteries

The **total backup service** provided by the battery over its service life ('P_{BU}'), expressed in kW·min·y, shall be calculated as follows:

$$P_{BU} = backup\ power\ capability\ \cdot\ years\ of\ operation\ = (P \cdot t_{se}) \cdot years\ of\ operation$$

Where:

- ***backup power capability*** ('$P \cdot t_{se}$') is the product of the power of the battery ('*P*'), expressed in kW, and its stored energy time ('t_{se}'), in minutes, available over the whole service life.
  - power of the battery ('*P*') refers to the Power (in kW) declared according to Article 10 / Annex IV of Regulation (EU) 2023/1542.
  - Stored energy time ('t_{se}') refers to the time (in minutes) the battery is able to provide its power *P* above the specified minimum voltage in any moment of its service life (except during recharge right after having provided backup service). The stored energy time t_{se} is equivalent to the '*final useable energy capacity*' divided by the power *P* of the battery (and finally converted in minutes):

$$t_{se} = \frac{final\ useable\ energy\ capacity\ [kWh]}{P\ [kW]} \cdot 60\ \left[\frac{min}{h}\right]$$

  - '*Final useable energy capacity*' refers to the energy available to the user when discharging a fully charged battery at its end of service life within the charge/discharge limits set by the battery management system under standard operating conditions.
  - "Fully charged" refers to charging until the charge level set by the battery management system under standard operating conditions.
- ***years of operation*** is the battery lifetime expressed in years, as declared according to Article 10 / Annex IV of Regulation (EU) 2023/1542, or the battery lifetime in years as specified in the technical specification delivered together with the battery, whatever is lower.

Note: Since the exact procedure of how the parameters of Article 10 / Annex IV are determined is not yet available, it might be possible that the lifetime under Article 10 differs from that given in the technical specification delivered with the battery e.g., because of different test conditions. To account for this possibility, the lower value shall be used. However, this option might prove irrelevant when the detailed procedure of how to determine the Article 10 parameters is published.

The total backup service ('P_{BU}'), the power ('*P*'), the stored energy time ('t_{se}'), the final useable energy capacity, the lifetime in years, the standard used for determining the battery service life (years of operation), the standard operating conditions, and the threshold for determining the end of service life shall be provided in the public version of the CFB supporting study. Their values shall correspond to those declared according to Article 10 / Annex IV of Regulation (EU) 2023/1542.

In case the corresponding documentation is not provided or if the above-mentioned requirements are not met, a default value of 2 years shall be used for lifetime in years.

Note: The term t_{se} is derived from IEC 62040-3, but it is of general applicability (based only on declared parameters, it does not imply any specific test procedure).

Note: The backup power capability might also be communicated in kWh if more appropriate (e.g., for long discharge OND batteries) as long as the corresponding conversion factor is applied for the calculation of P_{BU}.

3.2.2.2 Custom- made (CM)- type OND industrial batteries

The **total backup service** provided by the battery over its service life ('P_{BU}'), expressed in kW·min·y, shall be calculated as follows:

$$P_{BU} = backup\ power\ capability\ \cdot\ years\ of\ operation\ = (P \cdot t_{se}) \cdot years\ of\ operation$$

Where:

- The ***backup power capability*** ('$P \cdot t_{se}$') is the product of the power of the battery ('*P*'), expressed in kW, and its stored energy time ('t_{se}'), in minutes, available over the whole service life, as specified in both the final technical purchasing specification (TPS) and the corresponding auditable technical specification (ATS) that are underlying the contractual agreement.
- ***years of operation*** is the battery lifetime expressed in years as specified in both the final technical purchasing specification (TPS) and the corresponding auditable technical specification (ATS) that are underlying the contractual agreement.

The *backup power capability* may be given not at constant discharge power but according to a specific discharge profile as provided in the TPS. In this case, the average power over the specified discharge time window shall be used for determining the *backup power capability*.

The TPS and the ATS shall be fully disclosed in the CFB supporting study, including the underlying application load profile. The corresponding auditable technical specification (ATS) delivered with the battery shall refer to the TPS and list all replacement parts that are needed so that the battery as delivered can fulfil the application requirements as provided in the TPS. The discharge power ('*P*') and discharge time ('t_{se}'), the final useable energy capacity, and the lifetime in years shall be provided in both the TPS and the ATS. In case of discrepancy between the values provided in the TPS and in the ATS for any of these parameters, the lower value shall be used. The values provided

in the ATS shall correspond to the values declared according to Article 10 / Annex IV of Regulation (EU) 2023/1542 and be disclosed in the public version of the CFB supporting study.
In case the corresponding documentation is not provided, or it does not fulfil the above-mentioned requirements, the battery shall be treated like an OtS battery.

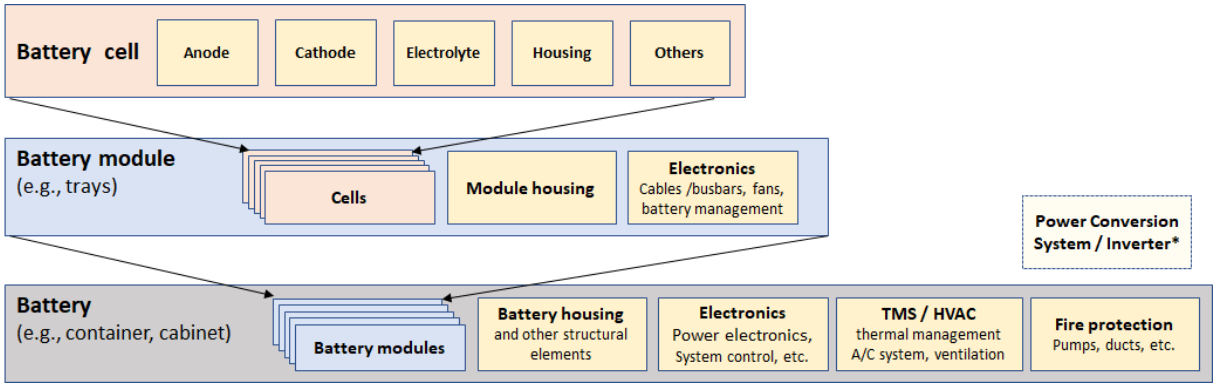
Note: The backup power capability might also be communicated in kWh if more appropriate (e.g., for long discharge OND batteries) as long as the corresponding conversion factor is applied for the calculation of P_{BU}.

4 System Boundaries and cut-off rules

The system boundaries define which parts of the product life cycle and which associated life cycle stages and processes belong to the analysed system (i.e., are required for carrying out its function as defined by the functional unit).

Batteries falling under the scope of this document can have very diverse layouts and configurations, given the wide range of applications they are designed for. Typically, several battery cells are grouped together to form battery modules, which are then mounted together in a specific structure to form an operative battery such as a cabinet-based stationary battery, a battery pack for mobile applications, a home storage battery, or a large container battery. A schematic representation of the considered battery system components is provided in Figure 1. The definition of the system boundaries follows a general supply-chain logic, including all stages from raw material acquisition and pre-processing, production of the main product, product distribution, etc.

Figure 1. Battery system components



*Source: JRC own elaboration. * Power Conversion System / Inverter excluded from the considered system components. TMS: Thermal Management System; A/C: Air Conditioning; HVAC: Heating, Ventilation and Air Conditioning.*

<u>Note</u>: The battery subject to the CFB declaration is the battery as placed on the market. Some system components may therefore not apply to certain batteries (e.g., no fire protection for OtS home storage systems). Battery is a generic concept and can refer to the final installation of the system at the site of operation or to the integration of the battery into the system / application where it is operated. As such, it may refer to a readily assembled OtS container system, but also to a battery that is assembled from its system components within the application environment (e.g., a train) and placed on the market as part of the application.

4.1 System boundaries

Whenever a physical system is necessary for the battery to safely and reliably deliver its performance and is dedicated to the use of this specific battery as specified in this paragraph, it shall be inside the system boundaries and be modelled according to the present rules.

- For **CM batteries**, the system boundaries comprise at least all components that are part of the battery (system) as delivered to the customer according to the corresponding auditable technical specification (ATS), including all replacement parts and components eventually required over the specified lifetime for fulfilling the requirements of the technical purchasing specification (TPS), and at least all components that are physically contained in or permanently attached to the battery housing. If the battery as delivered to the customer deviates for any reason from the TPS underlying the contractual agreement or if any amendments are made to the contractual agreement in this regard, these shall be considered for the CFB declaration and documented appropriately in the CFB supporting study.
  If a CM battery requires external components for its safe and reliable operation or for delivering its performance as specified in the TPS (such as a charge control system or an external battery management system) or if it requires services provided by components that are part of the application (such as the input of coolant at a certain temperature from the application's air conditioning system), then these components shall be included in the system boundaries and accounted for in the CFB calculation, applying (if applicable) the allocation hierarchy according to Section 6.2.1. Similarly, if a battery is installed in an existing premise and substantial modifications are needed for ensuring the battery's safe and reliable operation as specified in the TPS (such as a retrofitting of the existing fire extinguishing system or an additional containment), these components fall within the system boundaries.
- For **OtS batteries**, the system boundaries comprise at least the battery as being sold on the market including all components contained within or permanently attached to the battery housing and as required for delivering the performance specified in the technical documentation provided with the battery and as declared in the CFB supporting study.

Being the essential service of a battery the provision of direct current (DC) or DC power, any subsequent conversion of the delivered DC electricity (such as alternating current (AC) conversion, transformation), may be excluded from the system boundaries, applying subdivision according to Section 6.2.1.

Note: IEC 63369 includes all the components needed for safe operation and proper functioning of the battery in the system boundaries, such as battery-related part of the charger (if it ensures some of the functions of the battery management system) or of the power conversion system (PCS) or of the fire suppression system (FSS) i.e., those that are required for functional safety of the battery. If some of these parts are incorporated in devices located outside of the battery, then the percentage of the external device performing the battery related function shall be allocated to the battery following the allocation hierarchy as specified in Section 6.2.1 *(subdivision or allocation based on physical relationship i.e., considering the percentage of the weight or surface area of the device allocated to the battery function). This allocation shall be detailed in the system boundaries and reported in the CFB supporting study.*

4.2 Life Cycle Stages

The life cycle stages that shall be included in the system boundary are depicted in Figure 2 and described in the following. Other types of batteries (e.g., anode-free batteries) may have a different visualization of their system boundaries. The squares in Figure 2 represent processes, while arrows represent inputs (activity data such as kg of solvent, kg of additive etc.). The different colours (blue, yellow, orange, and green) indicate to which life cycle stage each process belongs, while red arrows and red borders indicate that a process/activity data shall be company-specific (Section 5). Black arrows and black borders indicate non-mandatory company-specific activity data and processes,

respectively. The use-phase of the battery is explicitly excluded from the system boundaries according to Article 7 and Annex II of Regulation (EU) 2023/1542.

4.2.1 Raw material acquisition and pre-processing

This life cycle stage covers all activities within the system boundaries prior to the main product production stage, including, but not limited to:

- the extraction of resources from nature and their pre-processing until their use in product components entering through the gate of the first facility falling under the main product production life cycle stage;
- transport of raw materials and intermediate products within, between and from extraction and pre-processing facilities until the first facility falling under the main product production life cycle stage;
- the production of all materials or intermediate products that are not covered by the main product production stage, such as the cathode active material precursors, anode active material precursors, electrolyte salts and solvents, electric and electronic equipment, the components of the thermal conditioning and fire suppression system.

All activities and elementary flows related to the management of the waste generated during this life cycle stage shall be included in calculation of the carbon footprint of this life cycle stage.

4.2.2 Main product production

The life cycle stage of the 'Main product production' covers the following activities:

- cathode active material production;
- anode active material production, including the production of graphite and hard carbon from their precursors;
- anode and cathode production, including the mixing of ink components, coating of ink on collectors, drying, calendaring, slitting, grid casting, grid pasting, curing and plate formation;
- electrolyte production, such as mixing of electrolyte salts and solvents;
- assembling the cell components into a battery cell, including stacking/winding of electrodes and separator, assembly of electrode plates with separator, assembling into a cell housing or pouch, injection of electrolyte, closing of cell, testing and electrical formation;
- assembling the cells into modules/pack including electric/electronic components, housing, and other relevant components that are necessary for the safe and reliable operation of the battery;
- assembling the modules/packs with electric/electronic components, housing, thermal conditioning (heating/cooling/HVAC), fire suppression system and other relevant components such as wiring into a finished battery;
- retrofitting activities if the battery is installed in existing premises that require a substantial modification for ensuring the battery's safe and reliable operation (such as a retrofitting of the existing fire extinguishing system or installing an additional containment). This applies to CM batteries in case such requirements are specified in the auditable technical specification (ATS) underlying the contractual agreement;
- transport operations of the intermediate products to the site where they are used.

The modelling of the management of the waste generated during this life cycle stage shall be included in calculation of the carbon footprint of this life cycle stage. More information on how to account for manufacturing scraps and recycled content in the 'Main product production' life cycle stage is provided in Section 6.3.2.

4.2.3 Distribution

The 'Distribution' life cycle stage covers the transport of the battery from the battery manufacturing site to the point of placing the battery on the market. Storage operations are not covered.

4.2.4 End of Life and recycling

The 'End of Life and recycling' life cycle stage begins when the battery or the system in which the battery is incorporated is disposed of or discarded by the user or the operator and ends when the battery concerned is returned to nature as a waste product or enters another product's life cycle as a recycled input. This life cycle stage covers at least the following activities:

- the transport of the waste batteries from the site of operation (stationary applications) or of vehicle dismantling (mobile applications) to the battery dismantling site;
- battery dismantling;
- thermal or mechanical treatment, such as electrolyte extraction and desulphurisation, thermal deactivation, pyrolysis, shredding of the waste battery cells, mechanical separation of the shredder fractions;
- battery cell recycling such as pyrometallurgical and hydrometallurgical treatment or smelting of previously separated metal fractions;
- separation and conversion into recycled material, such as recycling of the aluminium from the casing;
- recycling of electric and electronic components, such as printed wiring board ('PWB') recycling;
- energy recovery and disposal of waste fractions.

Note: populated printed wiring board (PWB) is used as a generic proxy for electronic components throughout this document.

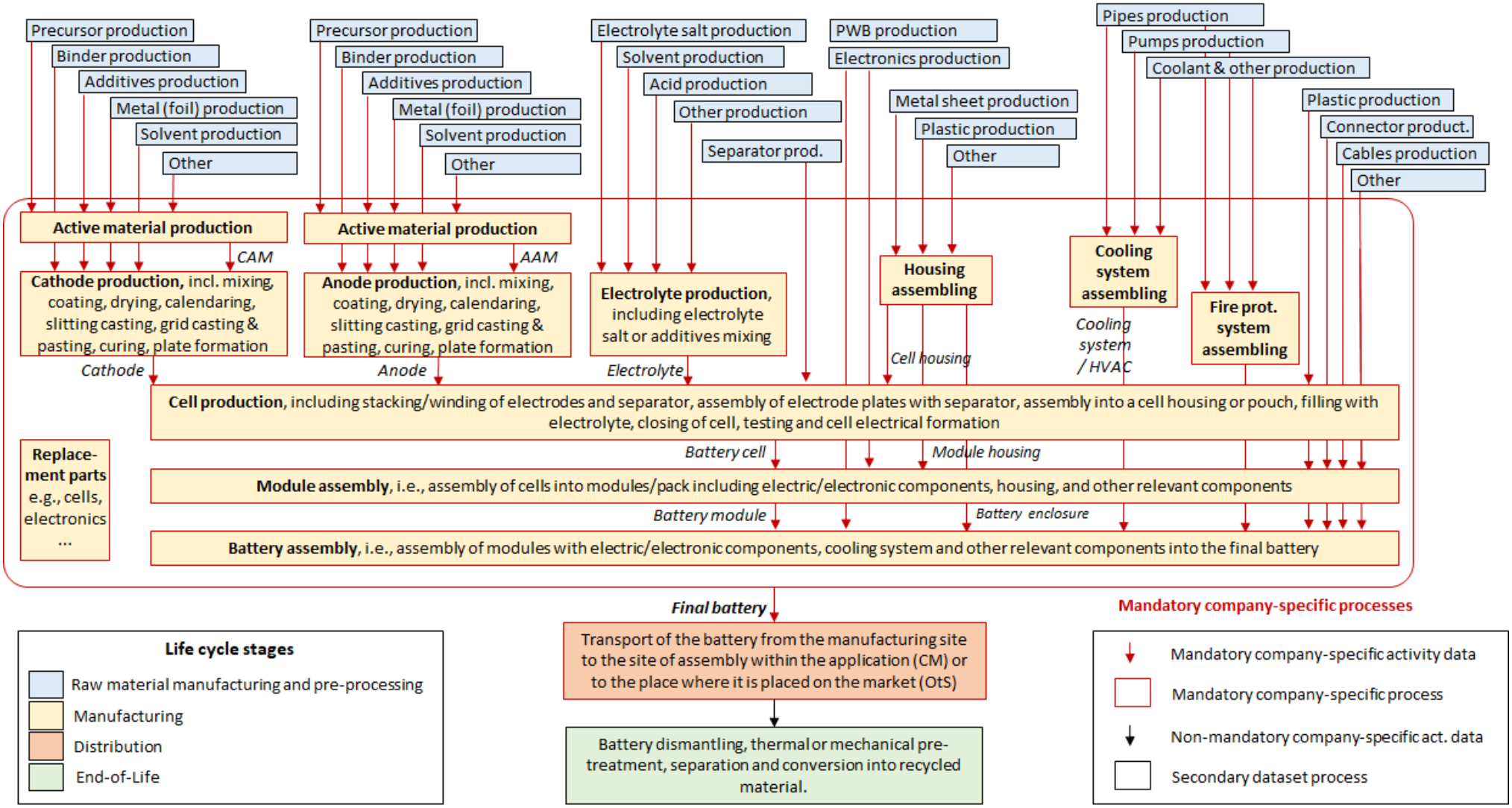
The environmental impacts of the pre-treatment of the waste batteries, such as decommissioning and disassembly of the battery (system), of discharging and sorting, and of the dismantling of the battery and its components are not included. However, any change in the composition of the product or of the corresponding mass balances (such as a separation process) shall be considered.

Note: Disassembly and sorting are mainly done manually and the environmental impacts of these activities are therefore negligible. In addition, considering these activities would create additional difficulties in the application of the CFF, which is to be applied per material, eventually requiring allocation.

Waste generated during the 'Raw material acquisition and pre-processing' and 'Main product production' life cycle stages shall be included in the life cycle of the product and modelled at the life cycle stage where it occurs. No repurposing, refurbishing or second life shall be considered in the CFB according to Article 7 of Regulation (EU) 2023/1542 i.e., the EoL stage begins with the end of the battery's service life in the primary application. A battery that is sent to repurposing,

refurbishing or second life shall be considered to enter the End of Life stage under the present framework, disregarding the lifetime and the service provided in the second life.

Figure 2. System boundaries for a generic industrial battery. Processes and materials may vary depending on the battery type and chemistry



Source: JRC own elaboration. PWB: populated Printed Wiring Board. CAM: cathode active material. AAM: anode active material. HVAC: Heating, Ventilation, and Air Conditioning.

4.3 Cut-off rules

The following processes may be excluded from the modelling:

(a) manufacturing of capital goods, including equipment;

(b) production of packaging materials;

(c) any component that is external to the battery as delivered to the client / the final user (such as the building where a battery is installed in) and that is not essential for the battery's safe and reliable operation;

(d) auxiliary inputs to manufacturing plants that are not directly related to the battery production process, including heating and lighting of associated office rooms, secondary services, sales processes, administrative and research departments;

(e) Regular maintenance activities that are not associated with the replacement of components or active materials, such as revisions, cleaning, or make-up of auxiliary substances like cooling / HVAC liquids, or fire extinguishing agents.

Note: the auxiliary inputs to the 'Main product production' life cycle stage were estimated to be negligible: assuming a consumption of a generic office as 293 kWh/m^2 (Pérez-Lombard, Ortiz and Pout, 2008) *and assuming a production area of 267000 m^2 (compatible with a facility of 20 GWh/year of production capacity), the auxiliary energy consumption is estimated to be about 0.2% of the electrical energy required by the production line.*

A general cut-off of 1% in mass may be applied to material inputs per system component as depicted in Figure 1, by neglecting input and output flows that make up less than 1% to the total mass of the system component.

The cut-off may be applied within the modelling of the production of the following system components in the 'Main product production' life cycle stage:

(a) battery cell anode;

(b) battery cell cathode;

(c) battery cell electrolyte;

(d) battery cell housing;

(e) the battery cell components other than the ones listed in points (a) to (d) combined;

(f) battery module housing;

(g) battery module electronics;

(h) battery module other than housing or electronics;

(i) battery housing / enclosure;

(j) battery electronics;

(k) battery fire protection and battery thermal conditioning system;

(l) battery other components than the ones listed in points (i) to (k) combined.

The cut-off may be applied within the modelling of the following processes in the 'Raw material acquisition and pre-processing' life cycle stage:

(a) mining;

(b) beneficiation or ore processing, from ore to concentrate;

(c) primary extraction, either pyrometallurgical or hydrometallurgical;

(d) refining;

(e) finishing.

Grinding media in the raw material acquisition and pre-processing life cycle stage shall be accounted for even if it might fall under the cut-off criteria.

If a cut-off is applied, the mass gap shall be closed on system component level (in the case of 'main product production' life cycle stage), or on process level (in the case of 'raw material acquisition and pre-processing' life cycle stage), by adding the missing mass to the material input flow with the highest specific carbon footprint among all remaining materials used in the system component to which the cut-off was applied. The carbon footprint study shall mention if and where the cut-off of 1% in mass has been applied.

5 Data collection requirements and quality requirements

A CFB model is composed of processes that combine elementary flows with the corresponding characterisation factors, and activity data with the corresponding life cycle inventory or carbon footprint of the underlying process.

In this document, life cycle inventory (LCI) refers to the combined set of exchanges of elementary, waste and product flows of a process, quantitatively describing all exchanges caused by the production of a given unit of the product. Activity data in turn represent the activities of a process that each are multiplied by the corresponding characterisation factors (elementary flows) or with the aggregated LCI result of each dataset used for modelling the corresponding exchange to derive the carbon footprint associated with that process (Annex II Regulation (EU) 2023/1542). Thus, the LCI data describes the actual process, while the activity data is linked with the impact assessment after allocation according to Sections 6.2 and 6.3.1.

Both data (i.e., activity data and elementary flows) from processes and the underlying sub-processes can be company-specific or secondary (i.e., taken from databases). The CFB Rules distinguish between mandatory company-specific processes (Section 5.1) and non-mandatory company-specific processes (Section 5.2).

All data shall be provided as LCI and activity data (i.e., before and after allocation), both per kg of the main output (product) of the modelled process and per functional unit.

5.1 Mandatory company-specific processes

The data collection and modelling for all the processes included in the main product production and the distribution life cycle stages shall be based on company-specific data. Guidelines on the collection of company-specific data are provided in Section 5.4.

The manufacturer of the battery shall ensure that the company-specific data is communicated in any of the following methods:

a) suppliers provide to the manufacturer the complete LCI of the process, including elementary flows, energy consumption, input material, and the recycled content ('R_1') referred to in Section 6.3, and the information required for the carbon footprint study as specified in Section 7;

b) suppliers provide the manufacturer with a company-specific dataset of the process;

c) suppliers provide the complete LCI of the process, including elementary flows, energy consumption, input material, and the recycled content ('R_1') referred to in Section 6.3, and the information required for the carbon footprint study as specified in Section 7 to a third-party, such as a data management company, who combines the inputs from different suppliers and provides a company-specific dataset for the process to the manufacturer.

Where the manufacturer communicates the company-specific data in accordance with point (b), the manufacturer shall ensure that the notified body receives from the manufacturer's suppliers all the information specified in Section 7 when the manufacturer lodges its application for assessment by the notified body. The manufacturer shall also ensure that a market surveillance authority receives such information upon request.

Where the manufacturer communicates the company-specific data in accordance with point (c), the manufacturer shall ensure that the notified body receives from its suppliers or from the third-party all the information specified in Section 7 when the manufacturer lodges its application for

assessment by the notified body. The manufacturer shall also ensure that a market surveillance authority receives such information upon request.

Company-specific datasets communicated by the suppliers to the manufacturer shall be accompanied with the information needed by the manufacturer or another of its suppliers to incorporate the company-specific dataset into its carbon footprint model and shall contain at least the following information:

a) a precise description of the product for which the carbon footprint information is valid, including its origin and an unambiguous identifier;

b) the total carbon footprint declared per unit of mass of the product, in kg CO_2-equivalent per kg, or per unit of energy, in kg CO_2-equivalent per kWh;

   Note: For intermediate datasets i.e., datasets for specific materials that are provided by suppliers to the manufacturer, their carbon footprint can be reported using the most appropriate functional unit. The requirements regarding the functional unit laid out in Section 3.2 applies to the final battery for which the CFB is declared.

c) the carbon footprint for each applicable life cycle stage listed in Section 4.2;

d) parameters related to the quality of the product that affect its carbon footprint, such as purity or specific capacity;

e) the specific content and the carbon footprint of elements and their compounds contained in the battery, at least regarding steel, aluminium, copper, cobalt, nickel, manganese, lithium, graphite, silicon, titanium, vanadium, silver, gold, platinum group metals, lead and phosphorous. If such materials contain recycled content, the carbon footprint of the specific emissions and resources consumed arising from the acquisition and pre-processing of primary material ('E_{V_Mat}') and the specific emissions and resources consumed arising from the recycling process of the recycled material ('$E_{recycled_Mat}$') as defined in Section 6.3.1 shall be provided;

f) in case a company-specific battery cell recycling process is modelled, the specific content of the materials targeted by the battery cell recycling process, their carbon footprint, their recycled content 'R_1', 'E_{V_Mat}' and '$E_{recycled_Mat}$' as specified in Section 6.3 shall be provided, if not covered by point e) and if requested by the battery manufacturer.

Note: The information about the content of these elements and their carbon footprint is required for the EoL modelling (Section 6.3.1), allowing to credit their specific carbon footprint when recovered at their EoL.

5.2 Non-mandatory company-specific processes

Non-mandatory processes are divided into most relevant and non-most relevant processes depending on their relevance on the overall CFB calculation. Guidelines on the modelling of most relevant non-mandatory company-specific processes are provided in Section 5.2.1, and on the modelling of non-most relevant ones in Section 5.2.2.

The most relevant processes are the following:

— production of the following cathode active material precursors: cobalt, nickel, iron, lithium, cadmium, lead, vanadium (all either metallic or as chemical compounds);

— production of the following anode active material precursors: lithium, titanium, nickel, cadmium, lead (all either metallic or as chemical compounds), rare earths, sodium metal, silicon, graphite and hard carbon precursors (such as natural graphite ore, cokes, etc.);

— production of the following electrolyte (salt) and precursors: Lithium hexafluorophosphate ($LiPF_6$), Lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), lithium compounds, sodium compounds, nickel compounds, solid electrolytes;

— production of copper, e.g., in the current collectors, busbars and cables;

— production of aluminium, e.g., in the current collectors, busbars / cables and housing;

— production of steel, e.g., in the housing.

Note: the most relevant processes include all that had been identified for EV batteries, plus several additional materials that have been deemed relevant for Industrial Batteries based on scientific literature and expert judgement. Examples of non-most relevant processes are the production of the solvent used in cathode and anode, of the carbon fibres used in the housing, of the electronics/PWB, the manufacturing of the plastic for the collector or housing, etc.

5.2.1 Modelling requirements of the most relevant processes

If at least one secondary dataset with a Technological Representativeness ('TeR') quality rating equal to or lower than four determined in accordance with Section 5.5 is available in the list of carbon footprint datasets dedicated to the carbon footprint of batteries on the European Platform on LCA ('the list of carbon footprint datasets') one of the following methods shall be chosen for data collection and modelling:

— the most representative secondary dataset in the list of carbon footprint datasets shall be used. If the secondary dataset is partially disaggregated, the electricity dataset or datasets connected to the core process one level down the supply chain, at -1 level, may be changed for the representative average electricity consumption mix of the country where the process is occurring, modelled in accordance with Section 6.1. Such choice shall be duly justified in the carbon footprint study;

— a company-specific dataset with a Data Quality Rating ('DQR') equal to or lower than 2.5. In such case, Section 5.1 shall apply.

If no secondary dataset with a TeR equal to or lower than four is available in the list of carbon footprint datasets, one of the following methods shall be chosen for data collection and modelling:

— a secondary dataset in line with the following hierarchy:

  - the most representative EF-compliant dataset commissioned by the European Commission available in LCDN. If the EF-compliant dataset is partially disaggregated, the electricity dataset or datasets connected to the core process one level down the supply chain, at -1 level, may be changed for the representative average electricity consumption mix of the country where the process is occurring, modelled in accordance with Section 6.1. Such choice shall be duly justified in the carbon footprint study;

  - a representative International Reference Life Cycle Data System (ILCD) entry-level compliant dataset.

— a company-specific dataset with a DQR equal to or lower than three. In such case, Section 5.1 shall apply.

For each process, the type of dataset selected shall be detailed in the carbon footprint study, including any relevant assumptions and justifications such as the choice of a proxy in the case of TeR equalling four.

Note: A company-specific dataset developed for a specific material produced by the company may be used, after being verified, by all customers that purchase this material for different CFB declarations.

5.2.2 Modelling requirements for the non- most relevant (other) processes

If one or more secondary datasets with a TeR quality rating equal to or lower than four determined in accordance with Section 5.5 are available in the list of carbon footprint datasets, the most representative secondary dataset in the list of carbon footprint datasets shall be used.

If no secondary dataset with a TeR quality rating equal to or lower than four is available in the list of carbon footprint datasets, a secondary dataset in line with the following hierarchy shall be used:

— the most representative EF-compliant dataset commissioned by the European Commission available in LCDN;

— a representative ILCD entry-level compliant dataset.

All the secondary datasets used shall be reported in the carbon footprint study.

5.3 CFB-compliant company-specific datasets

Company-specific datasets shall comply with the following requirements:

— the modelling shall be done in accordance with the rules set in this document;

— the data format shall be the ILCD data format available in LCDN;

— the nomenclature of the elementary flows shall be aligned with the EF 3.1 reference package for the carbon footprint of batteries available in LCDN;

— the name fields of the company-specific dataset and of the product flows shall be compliant with the ILCD Handbook Nomenclature and other conventions, available via the European Platform on LCA;

— they shall include the DQR and the values of the two DQR criteria, calculated in accordance with Section 5.5;

— the meta-data information shall comply with the requirements for meta-data information set out in the Guide for EF-compliant datasets, available on the European Platform on LCA[4] (European Commission: Joint Research Centre, European Platform on LCA (EPLCA), 2022);

— the system boundaries of cradle-to-gate models shall not include the distribution and EoL and recycling life cycle stages and only the material input of the circular footprint formula referred to in Section 6.3.1 shall apply;

— multi-functionality has been addressed in accordance with Section 6.2;

— they shall be externally verified.

[4] https://eplca.jrc.ec.europa.eu/LCDN/developerEF.html

5.4 Requirements for the collection of company-specific Life Cycle Inventory (LCI) data

The company-specific data to be collected for the creation of company-specific datasets shall include the inputs and outputs for the processes concerned, including:

- the following inputs:
  - material inputs that end up in the product, including minerals and metals, semi-finished materials and chemical feedstock. If materials are used in solution state, the specific concentration shall be provided. The specific concentration data on the metal and on other elements, either concentration or specific metal content shall be provided;
  - energy that is consumed directly and indirectly in the processing plant, such as electricity, steam, thermal energy required by the process, and energy and fuels required for auxiliary activities such as transport or forklifting within the plant premises;
  - auxiliaries and any other material inputs required for the manufacturing process, such as chemicals, cleaning material, lubricants, and refrigerants;
  - transport distances and means of transport;
  - any elementary flow;
- the following outputs:
  - any material output, including wastewater;
  - any elementary flow. Emissions that are not accounted for in the corresponding energy process dataset and that are not monitored via measurements shall be estimated based on stoichiometric calculations.

Company-specific data shall be the average of one year. However, the data may be the average of a different period if the process concerned has not yet been running for a full year or exceptionally in another case justified in the carbon footprint study.

A production process may be divided into sub-processes. The company-specific data may be collected for each process or sub-process stage separately, or for the final production as a whole. For the outputs, direct emissions and waste streams shall be recorded. For the inputs and outputs, the following information shall be recorded:

a) specification of the input or output, such as 'cobalt sulphate ($CoSO_4$ x7H_2O), primary, [provider], [country]';

b) unit, such as 'kg';

c) LCI data per kg main output product, before applying the circular footprint formula ('CFF') referred to in Section 6.3.1;

d) LCI data source, collection method or methods, and timespan;

e) LCI data collection date.

Measurements shall be collected at the points of consumption or emission directly relative to the process considered for the battery in the scope. The consumption of energy and auxiliaries shall preferably be based on an individual and detailed metering system that enables to attribute the energy or auxiliary consumption of the entire production to production lines, products, and time periods. Where the energy or auxiliary consumption cannot directly be related to a specific product,

for example where several products are produced in a facility while the consumption data is not available per specific product, the data shall be collected as specifically as possible, such as split up into energy or auxiliary consumption for electrode manufacturing, cell assembly, cell finishing, and climatization of clean or dry rooms. Where the energy or auxiliary consumption can be directly related to a specific process, the data of such consumption shall be used. Where the consumption data is only available for several products, for example, in case of presence of individual meters for cell assembly lines and only one general meter for a dry room in which several assembly lines produce different cells, the consumption data shall be split up by allocation in accordance with Section 6.2.2. Where the process concerns a new facility, extension of capacity or exchange of entire production line, up to six of the initial months may be excluded from the data collection.

Where LCI data for material inputs are not available for a specific cell or battery, but only available aggregated for several cell products, the cell mass balance may be used to determine the material inputs. In this case, the resulting bill of material shall include all facility-specific yields, such as manufacturing scrap and manufacturing waste rates, including entire cells at the end of line testing. The corresponding yield rates shall be provided in the carbon footprint study.

Company-specific elementary flows such as emissions may be based on direct measurements or be calculated combining company-specific LCI data, such as kilograms of fuel consumption in a boiler, with related emission factors from established sources, such as a specific emission factor for combustion of that fuel. If the process concerned is covered by EU emission trading system monitoring rules, the quantification requirements as set out in Commission Implementing Regulation (EU) 2018/2066 (European Commission, 2018) shall apply for the processes and greenhouse gases covered therein. The data may be scaled, aggregated or undergo other forms of mathematical treatment in order to bring them in line with the reference flow of the process.

Direct emissions such as process emissions of CO_2 and other greenhouse gas emissions from chemical reactions shall be quantified from the reaction stoichiometry. If acids are obtained as co-product from emission abatement, subdivision shall be applied as referred to in Section 6.2.1.

All data sources and mathematical treatments applied to the data shall be provided in the carbon footprint study.

A company quality management system such as in accordance with ISO 9001:2015, ISO 14001:2015 or Regulation (EC) No 1221/2009 of the European Parliament and the Council (European Union, 2009), or equivalent shall be applied in order to demonstrate that all the activity data have been collected and managed correctly, and that they are representative of a yearly average or other period referred to of the process in scope.

The template in Table 1 may be used for the company-specific LCI data collection for each of the process stages in the raw material acquisition and pre-processing life cycle stage.

Table 1. Generic data collection template for the raw material acquisition and pre-processing life cycle stage

Material	**Unit**	**Data**	**Specification**
Inputs			
Main input (ore, matte, precursor, etc.)			Not applicable in case of company-specific mining
Electricity			To be modelled in accordance with Section 6.1
Fuels for transport and machinery			Such as diesel, liquified natural gas (LNG) or hydrogen
Fuels for (process) heat generation			Such as natural gas, coal / hydrogen
External heat or steam supply			If heat or steam is sourced externally. The fuel shall be specified
Explosives			
Filling or structural material for production			Such as cement for backfilling (mining activities)
Acids			Such as sulphuric acid for acid leaching
Sulphur or H_2S			Such as for on-site sulphuric acid production
Neutralizer or slagging agents			Such as lime, limestone, NaOH or MgO
Electrodes			Such as graphite electrodes
Reductants			Such as coal, charcoal or hydrogen
Chemicals			Such as solvents, additives and any other chemical product required for the production process. For mining activities: bulk chemicals such as frother, dispersants or flocculants. Other chemicals may be aggregated and added to the major bulk chemicals
Tyres			Such as for specific machines at the mining stage
Technical gases			Such as nitrogen or oxygen, if purchased externally
Grinding media			Such as high strength steel balls or rods
Transport of input materials			Such as by truck, train, barge. Standard distances may be used
Other			Any other input that does not fall under the cut-off

Outputs			
Main product			Such as ore mined, matte, concentrated ore, final metal or metal salt, graphite ore, intermediate product. Assay data to be provided for specific metal contents or concentrations of ores or minerals and intermediate products
Co-products			Such as sulphuric acid or other metals or metal compounds than the main product. Allocation in accordance with Section 6.2.1
Waste			Including waste rock and any other waste flow to be treated or disposed of, either liquid or solid
CO_2 (fossil) and other GHG emissions			Based on fuels and explosives if combustion emissions are not considered in the corresponding fuel or energy dataset, reductants, electrodes and other reactions where GHG emissions occur, such as neutralization or precipitation with limestone

Source: JRC own elaboration

5.5 Data quality rating (DQR) of the CFB declaration and of CFB-compliant datasets

A Data Quality Rating ('DQR') shall be calculated for the declared value of the carbon footprint in accordance with the following procedure:

a. determine the quality rating for each of the two DQR criteria of Technological Representativeness ('TeR') and Geographical Representativeness ('GeR') for all the company-specific and secondary datasets used in the model in accordance with Table 2; For elementary flows such as direct emissions, the value of the two DQR criteria shall be one.

b. calculate the carbon footprint of each process by multiplying the carbon footprint of the dataset by the corresponding activity data;

   Note: As an example, if a process consumes 3 kWh of electricity and the carbon footprint of the dataset modelling the electricity consumption is equal to 0.6 kg CO_2eq / kWh, then 0.6 kg CO_2eq / kWh is multiplied by the corresponding activity data (3 kWh in this case) to obtain the absolute value of the carbon footprint of the process.

c. calculate the carbon footprint of each elementary flow by multiplying the characterisation factor of the substance by the elementary flow;

d. calculate the 'carbon footprint contribution', expressed in percentage, of each process and elementary flow. The carbon footprint contribution is the ratio between the individual carbon footprint of the process, or of the elementary flow, divided by the sum of the values of all processes and elementary flows. In case a process has a negative carbon footprint, take the absolute value for this process rather than the negative value, including in the denominator;

e. calculate the value of each DQR criterion of the declared value of the carbon footprint as a weighted average of the quality ratings of the DQR criterion concerned, weighted by the carbon footprint contribution of each process determined in point (d);

f. calculate the DQR of the declared value of the carbon footprint as the sum of the values of the two DQR criteria divided by two: (TeR + GeR)/2.

The DQR and the values of TeR and GeR of the carbon footprint shall be provided in the public version of the carbon footprint study.

The DQR of company-specific datasets shall be calculated with the following procedure:

a. determine the quality rating for each of the two DQR criteria of TeR and GeR for all the company-specific and secondary datasets used in the model of the company-specific dataset concerned in accordance with Table 2. For elementary flows such as direct emissions, the value of each of the two DQR criteria shall be equal to one;

b. calculate the carbon footprint of each process by multiplying the carbon footprint of the dataset by the corresponding activity data;

c. calculate the carbon footprint of each elementary flow by multiplying the characterisation factor of the substance by the elementary flow;

d. calculate the carbon footprint contribution, expressed in percentage, of each process and elementary flow. The carbon footprint contribution is the ratio between the carbon footprint of the process, or of the elementary flow, divided by the sum of the values of all processes and elementary flows. In case a process has a negative carbon footprint, take the absolute value for this process rather than the negative value, including in the denominator;

e. calculate the value of each DQR criterion of the declared value of the carbon footprint of the company-specific dataset concerned as a weighted average of the quality ratings of the DQR criterion concerned, weighted by the carbon footprint contribution of each process determined in point (d);

f. calculate the DQR of the declared value of the carbon footprint of the company-specific dataset concerned as the sum of the values of the two DQR criteria divided by two.

Table 2. Evaluation of the DQR criteria

Quality rating	$TeR_{dataset}$	$GeR_{dataset}$
1	The technology concerned is the same as the one in scope of the dataset.	The process modelled takes place in the country for which the dataset is valid.
2	The technology concerned is included in the mix of technologies in scope of the dataset, yet with some limited differences in the production pathways.	The process modelled takes place in the geographical region for which the dataset is valid.
3	The technology concerned is included in the dataset, with significant differences in the production pathway.	The process modelled takes place in one of the geographical regions where the dataset is valid for, such as in the case of a global dataset.
4	The technology concerned is similar, including in terms of systems boundaries and carbon footprint, to the modelled technologies in the dataset, meaning a technological proxy.	The process modelled takes place in a country that is not included in the geographical region or regions for which the dataset is valid, but it is estimated that there are sufficient similarities based on expert judgement.
5	In all other cases not listed in 1-4.	In all other cases not listed in 1-4.

Source: JRC own elaboration

Where chemicals are aggregated and added to the major bulk chemicals, as referred to in Table 1 (Section 5.4), the TeR is equal to 4.

If the electricity dataset or datasets connected to the core process at '-1 level' are changed for the representative average electricity consumption mix, the GeR of the dataset shall be calculated as follows:

$$GeR = GeR_{original} - \left(GeR_{original} - GeR_{modified,-1}\right) \cdot Contribution_{original,-1}$$

Where:

- $GeR_{original}$ is the GeR of the secondary dataset before changing the dataset describing the electricity consumption in the –1 level, in accordance with Table 2;
- $GeR_{modified,\ -1}$ is the GeR of the secondary dataset describing the electricity consumption in the '-1 level' after the adjustment;
- $Contribution_{original,-1}$ is the contribution, expressed as a percentage, of the carbon footprint impact of the electricity consumption in the '-1 level' compared to the total carbon footprint of the secondary dataset.

  <u>Note</u>: Example for GeR is adjustment if the electricity in the '–1 level' is changed to model a most relevant process in a company-specific dataset: The dataset is assumed to be the global production of a material, having a $GeR_{original}$ of 3 (Table 2). The GeR of the adjusted '-1 level' process (i.e., $GeR_{modified,-1}$) is assumed to be 1 as the electricity of the '-1 level' has been adjusted with the effective country-specific electricity consumption mix of the country where the process occurs (see Table **2***). The contribution of the electricity (i.e., $Contribution_{original,-1}$) to the total carbon footprint of this secondary dataset is, for instance, 33%. This means that the GeR of the secondary dataset after the electricity adjustment will be equal to: 3-(3-1)*33%=2.33.*

  This approach leads to a weighted adjustment taking into account the contribution of the '-1 level' adjusted datasets to the overall carbon footprint of the dataset. If e.g., the electricity contributes 100% to the total carbon footprint of the dataset, adjusting it to GeR=1 will lift the

secondary dataset to a GeR of 1. If the contribution of the electricity to the overall carbon footprint of the secondary dataset was negligible, also the impact of its –1 level adjustment to the GeR would be negligible.

6 Inventory data - Modelling requirements

6.1 Electricity modelling

It is recommended that the electricity modelling shall finally align with the requirements laid out in the final DA-EV defining the methodology for calculating the CFB-EV (still under finalisation at the date of the publication of the present report).

6.2 Allocation rules

6.2.1 Allocation of multi-functional processes

If a process or facility provides more than one function, that is, it delivers several goods or services ('co-products'), all inputs and emissions linked to the process shall be partitioned between the product of interest and the other co-products in line with the following hierarchy, with specific exceptions and requirements as detailed in this section:

a. subdivision, by splitting up the process into sub-processes that can be assigned unambiguously to one single product flow;

b. allocation based on a relevant underlying physical relationship such as mass or energy. This shall be based on the properties that most closely represent the drivers for the corresponding input, and duly be justified and documented in the carbon footprint study;

c. economic allocation based on the economic value of the different outputs. Economic value refers to the product of the physical amount of the considered output (in terms of mass, volume, energy etc.) and its corresponding price (€/kg, €/l, €/MJ, etc.).

By way of derogation from the first paragraph, economic allocation shall always be applied when the ratio between the economic values of at least two of the different outputs is higher than a factor of five. Such value differences shall be calculated based on a 10-year global price average for metals, ores and metal compounds, and five-year global price averages for all other commodities. A shorter timespan may be used if evidence is provided that the global prices are not available for 10 years and 5 years, respectively.

The economic value of the different outputs used for economic allocation shall be calculated based on the following data:

a. 10-year global price averages for metals, ores and metal compounds;

b. five-year global price averages for all other commodities;

c. five-years average for plant-specific revenues;

d. five-year average process costs for recycling and waste treatment processes.

All allocation factors, the approach for calculating them and the underlying data sources shall be disclosed in the carbon footprint study. If the global prices are not available for 10 years and 5 years, respectively, a shorter time span may be used if evidence is provided of such non-availability. If the process has been operative for shorter time than five years, a shorter time span may be used to determine revenues or process costs if duly justified in the carbon footprint study. If shorter time spans are used for global prices, revenues, or process costs, the carbon footprint calculation shall be updated at the latest when sixty months average values are available.

6.2.2 Allocation of energy and auxiliary inputs of production lines

If company-specific data is collected for energy or auxiliary inputs or other consumables from a plant where subdivision is not possible among the different production lines because only one monitoring device or energy meter is installed for several production lines, allocation may be applied if the production steps, production equipment, and the products themselves are similar, such as for battery cells with the same geometry, but with different properties.

In such case, the following hierarchy shall be used:

a) allocation by mass or other physical properties that most closely represent the drivers for the corresponding input. For cell manufacturing processes, allocation by mass may only be applied if all cell products have the same geometry, such as pouch, cylindrical or prismatic, and the same size, such as 'cylindrical 21700';

b) allocation using the installed production capacity or another appropriate criterion. For cell manufacturing processes, the production capacity shall be measured in cell energy capacity expressed in kWh.

The applied method shall be reported and documented in the carbon footprint study, including the justification thereof. In all cases, the sum of the allocation shall equal the total measured consumption.

6.3 Recycled content and End of Life modelling

The recycled content and the waste generated during all the life cycle stages shall be modelled with the use of the circular footprint formula ('CFF') and shall be reported at the life cycle stage where the waste generation occurs. This section describes how to apply the CFF to the recycled content and to the End of Life and recycling life cycle stage of the battery in scope. For any other case, Section 4.4.8 of Annex I to Commission Recommendation (EU) 2021/2279 shall apply (European Commission, 2021).

The CFF shall not be used for any materials or objects rejected during the battery manufacturing process that are re-used as an integral part in the same process and that do not need to be recycled, such as run-around scrap. However, emissions and process inputs associated with their processing shall be accounted for in the corresponding life cycle stage.

The End of Life and recycling life cycle stage of the battery shall be modelled either:

- with the default battery recycling process specified in this section, using the default CFF parameters provided in Table 3 and the default battery cell recycling process provided in Table 4 to Table 9;
- with a company-specific battery cell recycling process, which may only be modelled for the share of the batteries for which evidence is provided that they will be recycled in a specific recycling plant operating at commercial scale and that the specific recycling process corresponds to the battery model for which the carbon footprint is declared. Such specific recycling process shall be modelled with company specific data in accordance with Section 5.1. Table 3 specifies which default values shall be replaced with company-specific values in such case. In case a company-specific battery cell recycling process is modelled, the deviations from the default battery recycling process and the corresponding evidence shall be detailed in the carbon footprint study. The justification for applying a company-specific battery cell recycling process shall be provided in the public version of the carbon footprint study.

6.3.1 The circular footprint formula ('CFF') applied to the battery in scope

The Circular Footprint Formula (CFF) describes the allocation of impacts and credits from the EoL handling of the battery. The following rules apply:

- The default recycled content of the materials in the battery in scope shall be zero unless evidence of the traceability throughout the supply-chain is provided in the CFB supporting study.
- The modelling requirements for the 'End of Life' life cycle stage depend on the battery chemistry of the battery in scope (i.e., the battery for which the CFB is developed and declared).
- It is assumed that only a small fraction of the batteries is not properly returned and treated under EU legislation[5]. By default (and unless declared differently), the properly returned batteries are assumed to be sent to (manual) disassembly / dismantling (Figure 3)
- The outputs of the dismantling are the battery cells and remaining battery components that are sent to individual recycling processes (steel and aluminium from the housing, copper, electronics/PWB for recovering gold, silver, copper and palladium, cables for recovering copper). The sorted plastic is sent to energy recovery and the rest to landfill. The dismantled battery cell is then sent to the default battery cell recycling process, which varies according to the battery cell chemistry:
    1. The default battery cell recycling process for lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature such as sealed pouch, round cell or prismatic cells, shall be modelled according to Table 4 and Table 5, assuming a combination of a pyrometallurgical and hydrometallurgical treatment (that recovers copper sulphate, nickel sulphate, and cobalt sulphate). NiCd cells are assumed to undergo an additional cadmium distillation stage before the combined pyrometallurgical and hydrometallurgical treatment, in order to also recover cadmium. This additional cadmium distillation stage shall be modelled according to Table 6.
    2. The default battery cell recycling process for lead-acid batteries shall be modelled according to Table 7 and Table 8, assuming a mechanical shredding of the cells / monoblocks and subsequent re-melting of the lead fraction.
    3. The default battery cell recycling process for Na-S high temperature batteries assumes a semi-manual disassembly of the cells under protective atmosphere and recovery of the sodium as sodium hydroxide and of the steel and aluminium contained in the cells as metal scraps and shall be modelled according to Table 9.
    4. For other battery chemistries that do not fall under any of the previous categories, no established battery cell recycling processes have been identified and therefore the default EoL scenario assumes no battery cell recycling (i.e. no material recovery from battery cells).

[5] Article 61 of Regulation (EU) 2023/1542 requires battery producers to "ensure that all waste batteries [..] are collected separately". However, this is only possible for batteries that become waste batteries within the EU, and not necessarily for all that have been made available on the market. The return rate considers that there is a small fraction of batteries of unknown whereabout (e.g., exported) that are not captured by Article 61 and therefore not recycled properly under EU Legislation.

The remaining fraction of the batteries is assumed to be not properly returned and treated under EU legislation (named as "non-properly returned" batteries in (Figure 3) and assumed to be roughly dismantled (with a process similar to that for properly returned batteries above, but without carrying out the recycling of the electronics/PWB), whereas battery cells are assumed to be landfilled.

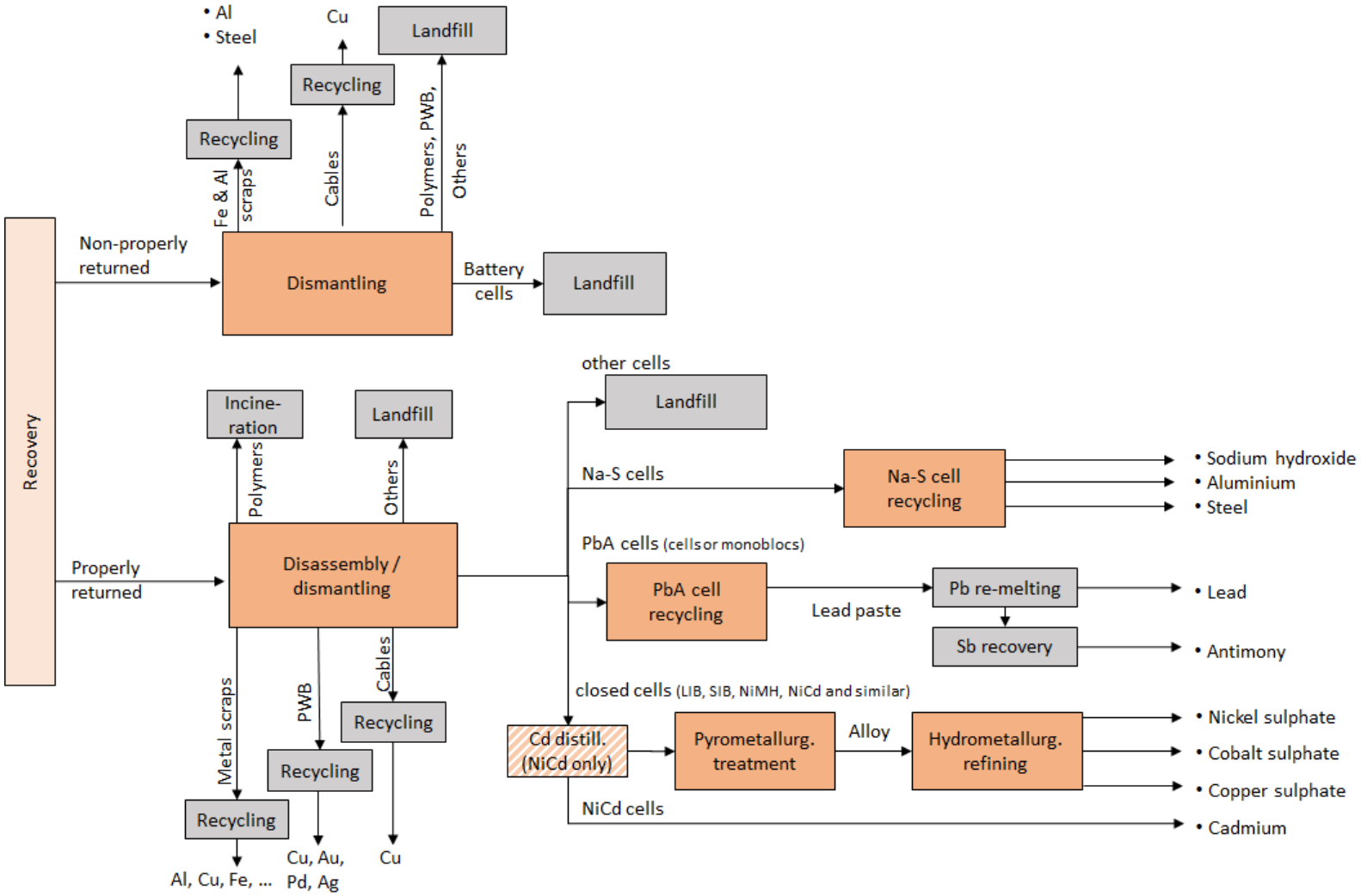
— A company-specific return rate may be used only for the share of batteries covered by an ownership business models where the property of the battery stays with the manufacturer and where contractual evidence of higher return rates is provided, including evidence that the batteries covered under that rate are recycled in accordance with Regulation (EU) 2023/1542.

— A company-specific battery recycling process may be used only for the share of the batteries for which the CFB declarant is able to provide evidence that the same batteries will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the CFB is declared. The corresponding evidence shall be provided in the CFB supporting study.

— The environmental impacts of collecting and dismantling the waste batteries shall not be included in the model (i.e., they are defined as cut-off in Section 4.3). The return rate is expressed with the parameters R_{Return} in the formula below, while the efficiency of the dismantling is assumed to be 100%.

— The incineration with energy recovery only applies to the plastic waste from the dismantling of the properly returned batteries. The energy recovery from plastics in the battery cells is instead accounted for by the default battery cell recycling process.

Figure 3. Schematic flowsheet of the default 'End of Life' life cycle stage of the battery in scope



Source: JRC own elaboration. PWB: populated Printed Wiring Board.

<u>Note</u>: The flowsheet in Figure 3 depicts the default recycling process, which reflects current industry practice. Other recycling processes may recover other materials (such as lithium). Such processes can be claimed and modelled if corresponding evidence is provided, including the corresponding credits for recovered materials.

The below equations describe how to apply the CFF to the battery in scope. The CFF shall be applied per material, unless specified differently in the description of the single parameters. In particular:

- the impacts of the battery cell recycling shall not be allocated per material but shall be allocated to the full battery using an average allocation factor ($A_{Batterycell}$) per battery cell;
- the impacts of recycling of the electronics/PWB shall not be allocated per material but shall be allocated to the full PWB using an average allocation factor (A_{PWB}) per the PWB.

The CFF is the sum of several terms, each related with different processes associated with the materials that constitute the battery in scope along its life cycle:

- Impacts of using primary and secondary materials ('material input'):
$$\sum_{Mat}[(1-R_{1_Mat})\cdot E_{V_Mat}+R_{1_Mat}\cdot(A_{Mat}\cdot E_{recycled_Mat}+(1-A_{Mat})\cdot E_{V_Mat}\cdot\frac{Q_{Sin_Mat}}{Q_{P_Mat}})]$$
- Impacts and credits of producing secondary materials from dismantling, such as steel and aluminium from the housing and copper from the cables ('dismantling'):
  - Materials from the properly returned fraction of the battery:
$$R_{Return}\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,r_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]$$
  - Materials from the non-properly returned fraction of the battery:
$$(1-R_{Return})\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,nr_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,nr_Mat}}{Q_{P_Mat}})]$$
- Impacts and credits of producing secondary materials from the PWB recycling after the battery dismantling, to account for copper, gold, silver and palladium ('electronics recycling'):
$$R_{Return}\cdot[(1-A_{PWB})\cdot E_{recEoL_PWB}-\sum_{Mat}[(1-A_{Mat})\cdot(R_{rec,r_Mat}\cdot E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]]$$
- Impacts and credits of producing secondary materials from the battery cell recycling, such as copper sulphate, nickel sulphate and cobalt sulphate in the default End of Life and recycling life cycle stage ('cell recycling') of LIB:
$$R_{Return}\cdot\left((1-A_{Batterycell})\cdot E_{recEoL_Batterycell}\right)+R_{Return}\cdot\sum_{Mat}[(1-A_{Mat})\cdot R_{rec,r_Mat}\cdot(E_{recEoL_Mat}-E^{*}_{V_Mat}\cdot\frac{Q_{Sout,r_Mat}}{Q_{P_Mat}})]$$

  <u>Note</u>: This term applies to any material targeted by any cell recycling process, and is therefore not limited to the materials mentioned exemplarily above for the default LIB cell recycling process. For the cell-chemistry-specific default recycling processes, the parameters given in Table 3 apply, including cadmium for NiCd, lead and antimony for PbA, and sodium hydroxide, aluminium and steel for Na-S. If an own company-specific cell recycling process

is modelled, it also applies to the corresponding recovered materials not listed in the default recycling process.

— Impacts due to energy recovery of the plastic from the dismantling of the properly returned battery waste ('energy recovery'):

$$R_{Return} \cdot \sum_{Mat} [(1 - B) \cdot R_{3,r_Mat} \cdot E_{ER_Mat}]$$

— Impacts of disposal ('disposal'):

  - PWB, polymers, battery cell, and other materials not-being recycled, deriving from the dismantling of the non-properly returned battery waste:

$$(1 - R_{Return}) \cdot \sum_{Mat} [(1 - R_{rec,nr_Mat}) \cdot E_{D_Mat}]$$

  - Other materials not-being recycled, deriving from the dismantling of the properly returned battery waste

$$R_{Return} \cdot \sum_{Mat} [(1 - R_{rec,r_Mat} - R_{3,r_Mat}) \cdot E_{D_Mat}]$$

Where:

a) A_{Mat} is the material-specific allocation factor of burdens and credits between two life cycles, the one supplying and the one using recycled material, aiming to reflect market realities. The values for A_{Mat} shall be taken from Table 3. In case a material is not included in Table 3, reference application-specific values shall be taken from Part C of Annex II to Commission Recommendation (EU) 2021/2279 (European Commission, 2021). If no such values for a specific application are there available, then material-specific values for A_{Mat} may be used from the same source. If such values of parameter A_{Mat} for the considered materials are not available either, a value of 0.5 shall be used;

b) $A_{Batterycell}$ is the battery-specific allocation factor and shall be equal to 0.2;

c) A_{PWB} is the PWB-specific allocation factor and shall be equal to 0.2;

d) B is the allocation factor of energy recovery processes and shall be equal to 0;

e) R_{1_Mat} is the material-specific recycled content, meaning the proportion of material in the input to the production that has been recycled from a previous system. R_{1_Mat} is by default equal to 0 for all the materials, unless evidence for a specific value is provided in the carbon footprint study, based on traceability of the supply-chain. Acceptable evidence shall include the documentation referred to in Article 8(1) of Regulation (EU) 2023/1542. Specific values based on supply market statistics are not acceptable;

f) R_{Return} is the battery return rate, meaning the rate of EoL batteries that are properly returned for removal of parts and components multiplied with a 100% collection of contained batteries to be recycled in accordance with Regulation (EU) 2023/1542. R_{Return} is equal to 0.95 by default, unless evidence is provided in the carbon footprint study for a different company-specific value. This may only be done for the share of batteries covered by an ownership business model where the property of the battery stays with the manufacturer and evidence of such ownership model shall be provided in the carbon footprint study together with evidence of the higher return rate, including evidence that the batteries covered under that rate will be recycled in accordance with Regulation (EU) 2023/1542;

<u>Note</u>: The return rate represents the percentage of industrial batteries that are recovered for recycling in Europe at the end of their life and that are treated with the corresponding recycling processes under EU legislation. The non-returned fraction is of unknown whereabout (e.g., batteries that leave the boundaries of the EU and that do not fall under the corresponding collection schemes).

g) R_{rec,r_Mat} is the material-specific recycling yield for the properly returned fraction of batteries, meaning the proportion of the material in the properly returned fraction of batteries that will be recycled in a subsequent system taking into consideration only the recycling yield. It does not include return rate and dismantling efficiency. Default values for R_{rec,r_Mat} are provided in Table 3. Different company-specific values may be used for R_{rec,r_Mat} in case a company-specific battery cell recycling process is applied and the corresponding evidence is provided in the carbon footprint supporting study. In case such company-specific values are applied, they shall be calculated as the ratio between the mass of the secondary material produced and the mass of the material entering the recycling process. In the case of copper, gold, silver or palladium from the electronics/PWB recycling, R_{rec,r_Mat} refers to the kg of such material per kg of PWB input and only the default values provided in Table 3 shall be used;

h) R_{rec,nr_Mat} is the material-specific recycling yield for the non-properly returned fraction of batteries, meaning the proportion of the material in the non-properly returned fraction of batteries that will be recycled in a subsequent system taking into consideration only the recycling yield. It does not include return rate and dismantling efficiency. The values for R_{rec,nr_Mat} shall be taken from Table 3;

i) R_{3,r_Mat} is the proportion of the material that is used for energy recovery at the EoL of the properly returned waste batteries. For polymers R_{3,r_Mat} shall be 100%. For other materials, R_{3,r_Mat} shall be 0 unless evidence for a different value is provided in the carbon footprint study;

j) E_{V_Mat} is the specific emissions and resources consumed arising from the acquisition and pre-processing of primary material;

k) $E_{recycled_Mat}$ is the specific emissions and resources consumed arising from the recycling process of the recycled material. Collection, sorting, and transportation of the waste used to produce the secondary material are excluded. The allocation of a multi-material process to the single material shall follow the allocation hierarchy set out in Section 6.2. It shall include the management of the waste of the recycling process and the wastewater treatment when applicable. In case $E_{recEoL_Batterycell}$ is modelled with a company-specific process taking place outside Europe, the geographical scope of $E_{recycled_Mat}$ data shall be changed accordingly.

<u>Note</u>: $E_{recycled_Mat}$ can be either a most relevant process or a non-most relevant process, depending on the material. The materials classified as most relevant are provided in Section 5.2.

l) $E_{recEoL_Batterycell}$ is the specific emissions and resources consumed arising from the battery cell recycling. $E_{recEoL_Batterycell}$ is accounted for on battery level and not per material. $E_{recEoL_Batterycell}$ shall be modelled using the default values provided in:

1. Table 4 and Table 5 for lithium-ion, sodium-ion, nickel metal hydride, nickel-cadmium batteries and similar battery chemistries that are based on closed battery cells with liquid electrolytes operating at room temperature (such as sealed pouch, round cell or prismatic cells);
2. Table 6 for NiCd cells for the additional cadmium distillation stage to be applied on top of the recycling process provided under point 1;
3. Table 7 and Table 8 for lead-acid batteries.;

4. Table 9 for sodium-sulphur high temperature battery cells;
5. For all other battery chemistries, no specific cell recycling process is available and the battery cells are assumed to be sent to landfill.

Different company-specific recycling processes may be used only for the share of the batteries for which contractual evidence is provided that they will be recycled in a specific recycling plant and that the specific recycling process corresponds to the battery model for which the carbon footprint is declared. In this case, the company-specific battery cell recycling process shall refer to the recycling plant operating at the time of the battery being placed on the market. The mass and composition of the battery used for the company-specific battery cell recycling process shall reflect the mass and composition of the battery for which the carbon footprint is declared. The justification for modelling a company-specific battery cell recycling process shall be provided in the public version of the carbon footprint study. In case a company-specific battery cell recycling process is applied, the options in Section 5.1 shall apply, to be read as applying to the recycler rather than the supplier. The default recycling process is assumed to take place in Europe, and the geographical region of Europe shall be used for modelling and selection of secondary datasets;

m) E_{recEoL_Mat} is the specific emissions and resources consumed arising from any additional recycling process needed to produce secondary material. In the default battery recycling process, data use shall reflect recycling processes taking place in Europe, both for the properly returned waste batteries and the non-properly returned waste batteries. In case $E_{recEoL_Batterycell}$ is modelled with a company-specific process taking place outside Europe, the geographical scope of E_{recEoL_Mat} data in the cell recycling shall be changed accordingly. For all the outputs of the battery cell recycling process, E_{recEoL_Mat} includes all the additional recycling processes not included in the battery cell recycling, such as sorting and re-melting of metal waste from the casing into secondary metals. It is equal to zero for the default battery cell recycling process since the obtained products in this default process do not require further post-processing;

<u>Note</u>: $E_{recEoL_Batterycell}$ includes all process inputs and direct emissions of the battery cell recycling process. E_{recEoL_Mat} are the specific burdens (carbon footprint) associated with the re-processing of the metal fractions separated during recycling, such as sorting and re-melting of aluminium scrap into secondary aluminium. E_{recEoL_Mat} is zero for all materials that leave the battery cell recycling process as final product i.e., directly replacing the equivalent primary product such as cobalt sulphate or nickel sulphate as in the default EoL life cycle stage. E_{recEoL_Mat} is not zero for any output of the battery cell recycling process that needs additional processes before being sold as secondary material. (e.g., sorting, cleaning and remelting to metal ingots).

n) E_{recEoL_PWB} is the specific emissions and resources consumed arising from the electronics/PWB recycling following the dismantling of the properly returned waste batteries. E_{recEoL_PWB} shall be determined in accordance with Section 5.2.2;

o) $E^*_{V_Mat}$ is the specific emissions and resources consumed arising from the acquisition and pre-processing of primary material assumed to be substituted by recyclable materials. $E^*_{V_Mat}$ shall be modelled based on the European average production or, when the European average production is not available, based on the global average production. However, $E^*_{V_Mat}$ shall be equal to E_{V_Mat} when E_{V_Mat} is lower than the applied average production;

p) Q_{P_Mat} is the quality of the primary material;

q) Q_{Sin_Mat} is the quality of the ingoing secondary material, meaning the quality of the recycled material at the point of substitution. Values for Q_{sin} / Q_p shall be taken from Table 3;

r) $Q_{Sout,r}$ and $Q_{Sout,nr}$ are the quality of the outgoing secondary material, meaning the quality of the recyclable material at the point of substitution, for the properly returned and the non-properly returned fraction, respectively. Default values for $Q_{Sout,r}$ / Q_p and $Q_{Sout,nr}$ / Q_p are provided in Table 3. Higher values are allowed only if a company-specific battery cell recycling process is modelled and if evidence is provided in the carbon footprint study, such as through technical specifications of the secondary material demonstrating that is 'battery-grade';

s) E_{ER_Mat} is the specific emissions and resources consumed arising from the energy recovery of the polymers from the battery waste dismantling, for both the properly returned and the non-properly returned waste batteries, including both the direct emissions of the incineration plants and the credits due to the avoided energy production. E_{ER_Mat} shall be determined in accordance with Section 5.2.2. If the secondary datasets for energy recovery do not include the credits from the production of energy, the credits shall be modelled using the lower heating value of the polymers and by substituting the European energy mix;

   <u>Note</u>: For this purpose, if incineration with energy recovery is modelled for such substances, check if the secondary dataset used for modelling the incineration process already includes the credits for recovered energy. If it does not, then the credit shall be modelled based on the lower heating value by crediting the corresponding amount of energy, using the dataset for the corresponding average European energy mix.

t) E_{D_Mat} is the specific emissions and resources consumed arising from the disposal of waste material at the analysed product's EoL, without energy recovery.

- If the datasets used for PWB recycling include the credits from the production of secondary material, the credits for specific materials shall be set to zero.
- If the used datasets already include the allocation with the factor A_{Mat}, then they shall not be allocated again.
- To apply the CFF correctly along the battery supply-chain, the company-specific R_1 of e.g., any raw material, precursors, active material, shall be handed over to the next process step / manufacturer even in case of CFB-compliant company-specific datasets.

  <u>Note</u>: When using an R_1 (recycled content) different from zero i.e., if secondary material is used as input for a process in the manufacturing life cycle stage, and if the secondary material is a product of a most relevant process as defined in Section 5.2, it may be modelled using company-specific data according to Section 5.2.1. In this case, the waste product (e.g., waste battery) that is processed by the recycling process for obtaining the secondary materials is burden free according to the EF Recommendation (2279/2021). The provider of the corresponding secondary material may then provide a company-specific dataset according to the present rules for the secondary material(s) obtained from its process. These datasets will be used as gate-to-gate processes that produce secondary materials ($E_{recycled_Mat}$).

  <u>Note</u>: If an own, company-specific recycling process is modelled where components are recycled on a higher level i.e., not broken down to individual materials but recovered as complex products, the CFF shall be applied per material. Complex product refers to components that are not recovered as individual materials (or elements), but on a higher level of complexity, such as in the case of re-use of components like battery trays, or for direct recycling of cathode active materials (CAM) where the CAM is not decomposed for recovering its constituents (cobalt

sulphate, nickel sulphate, etc.), but recovered integrally for re-use in the production of new battery cathodes. In this case, on the output side, credit may be given by material for $E^*_{V_Mat} = E_{V_Mat} + E_{gtg\text{-}c}$ *i.e., for the avoided primary material plus the corresponding gate-to-gate impacts of component manufacturing* ($E_{gtg\text{-}c}$)*. For the material-specific application of the CFF on the input side, a dataset shall be provided by the entity that produces the recycled complex product (such as the recycled CAM) for each of the materials that compose the complex product, giving the material-specific* $E_{recycled_Mat}$*. Allocation of the recycling process burdens* (E_{recEoL_Mat}) *to each of the materials that constitute the recycled complex product shall be done according to Section 6.2.*

Table 3. Default values for the parameters for the EoL and recycling life cycle stage

				For the properly returned waste batteries		For the non-properly returned waste batteries	
	A_{Mat}	R_{1_Mat}	Q_{Sin}/Q_p	R_{rec,r_Mat}	$Q_{Sout,r}/Q_p$	R_{rec,nr_Mat}	$Q_{Sout,nr}/Q_p$
Al metal (from dismantling)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0.9	1
Al metal (cells)	0.2	0 [(a)]	1	0 [(b)]	1 [(b)]	0	1
Al metal (in Na high temp cells)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0	1
Antimony	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0	1
Cd metal (cells)	0.2	0 [(a)]	1	0.99 [(b, f)]	1 [(b)]	0	1
Cu metal (from dismantling)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0.9	1
Cu metal (cells)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0	1
Fe metal (from dismantling)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0.9	1
Fe metal (cells)	0.2	0 [(a)]	1	0 [(b)]	1 [(b)]	0	1
Lead	0.2	0 [(a)]	1	0.9 [(b)]	0.97 [(e)]	0.8	0.9
Sodium metal (in Na-S cells)	0.2	0 [(a)]	1	0.9 [(b)]	1 [(b)]	0	1
Polymers (from dismantling)	0.5	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	0.8
Other materials (from dismantling)	0.5	0 [(a)]	1	0 [(b)]	n/a [(b)]	0	n/a
Au from PWB	0.2	0 [(a)]	1	1.4×10^{-5} [(c)]	1	0	1
Cu from PWB	0.2	0 [(a)]	1	0.11 [(c)]	1	0	1
Ag from PWB	0.2	0 [(a)]	1	9.77×10^{-4} [(c)]	1	0	1
Pd from PWB	0.2	0 [(a)]	1	9.31×10^{-8} [(c)]	1	0	1
Co compounds (cell)	0.2	0 [(a)]	1	0.9 [(b)]	0.8 [(b)]	0	0.8
Ni compounds (cell)	0.2	0 [(a)]	1	0.9 [(b)]	0.8 [(b)]	0	n/a
Mn compounds (cell)	0.2	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	n/a
Li compounds (cell)	0.2	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	n/a
Na compounds (cell)	0.2	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	n/a
Tin	0.2	0 [(a)]	1	0 [(b)]	1 [(b)]	0	n/a
Other metals and metal compounds	0.2	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	n/a
Graphite / hard carbon (cell)	0.2	0 [(a)]	1	0 [(b)]	0.8 [(b)]	0	n/a
Other materials (cell)	0.5 [(d)]	0 [(a)]	1 [(d)]	0 [(b)]	0.8 [(b)]	0	n/a

(a) Company-specific values may be used under the conditions set out in this section.
(b) Company-specific values shall be used if a company-specific battery cell recycling is used.
(c) Value shall be applied per kg of recycled electronics/PWB.
(d) Additional materials may be accounted for if a company-specific battery cell recycling process is modelled.
(e) The value $Q_{Sout,r}/Q_P$ for lead is due to the fact that PbA recycling obtains two products, highly pure lead and a lead alloy of lower lead content. The Q factor of 0.97 is estimated from the mass shares and the quality of each of both fractions (since primary lead is credited for both fractions in the EoL model).
(f) The value R_{rec,r_Mat} for cadmium is due to the fact that cadmium needs to be recovered with high efficiency due to environmental legislation.
n/a = not applicable.

Source: JRC own elaboration

Table 4. Default cell recycling model for recycling of LIB, NiMH, SIB and other battery cells of similar layout, inventory data per kg of battery cell, pyrometallurgical treatment. Equal to the CFB-EV

Component	Unit	Default value	Specification / dataset
Inputs			
EoL battery cells	kg	1.00	From disassembly
Electricity	kWh	1.24	Electricity mix
Process heat	MJ	0.360	Thermal energy from natural gas
Process heat	MJ	0.272	Thermal energy from diesel fuel
Limestone washed	kg	0.150	Limestone production, technology mix
Silica sand	kg	0.186	Silica sand, production mix
Quicklime (CAO)	kg	0.111	Quicklime production, technology mix
Wet lime ($Ca(OH)_2$)	kg	0.010	Calcium hydroxide production; technology mix
Carbon black	kg	0.001	Carbon black production, technology mix
Water	kg	0.298	Tap water, average technology mix
Transport from the disassembling to the battery cell recycling plant	km	130	Transport by truck (>32 t, EURO 4)
Transport from the disassembling to the battery cell recycling plant	km	240	Transport by train (average freight train)
Transport from the disassembling to the battery cell recycling plant	km	270	Transport by ship (barge)
Outputs			
Metal alloy	kg	0.209	To hydrometallurgical treatment
Slag	kg	0.952	Landfill of inert slag
Emissions	kg CO_2-eq.	1.051	Direct process emissions (all greenhouse gases)

Source: JRC analysis, based on PEFCR on rechargeable Batteries (RECHARGE, 2024).

Table 5. Default recycling model for recycling of LIB, NiMH, SIB and other battery cells of similar layout, inventory data per kg of battery cell, hydrometallurgical treatment

Component	Unit	Default value	Specification / dataset
Inputs			
Metal alloy	kg	0.209	
Electricity	kWh	2.59	Electricity mix
Process heat	MJ	4.79	Thermal energy from natural gas
Hydrochloric acid (100%)	kg	0.008	HCl production, technology mix
Hydrogen peroxide (100%)	kg	0.274	Hydrogen peroxide, production, technology mix
Soda (sodium carbonate)	kg	0.017	Soda production, technology mix
Sodium hydroxide (100%; caustic soda)	kg	0.42	Sodium hydroxide production, technology mix
Sulphuric acid aq. (96%)	kg	0.84	Sulphuric acid production, technology mix
Water (tap water)	m^3	0.01	
Outputs			
Recovered metals such as aluminium, copper, steel and stainless steel	kg		Credit for each specific metal according to its content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat}, and $Q_{Sout,r}/Q_p$
Recovered metal salts such as cobalt sulphate and nickel sulphate	kg		Credit for each specific metal salt according to its stoichiometric content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat} and $Q_{Sout,r}/Q_p$
Wastewater	m^3	8.64×10^{-3}	Wastewater treatment, average

Source: JRC analysis, based on PEFCR on rechargeable Batteries (RECHARGE, 2024)

Table 6. Default recycling model for NiCd battery cells, inventory data per kg of battery cell, distillation of cadmium

Component	Unit	Default value	Specification / dataset
Inputs			
EoL battery cells	kg	1	
Electricity	kWh	0.014	Electricity mix
Process heat	MJ	0.61	Thermal energy from natural gas
Natural gas	MJ	1.2	Direct input to pyrolysis process
Water (tap water)	m^3	0.01	
Outputs			
Recovered cadmium metal	kg		Credit for recovered metal according to its content in the battery cell and CFF parameters
Direct emissions	kg CO_2-eq.	0.083	From oxidation of natural gas (oxidation of other organic components in the cells is accounted for in the pyrometallurgical step)

Cadmium distillation is an additional process step for NiCd battery cells; to be applied in addition to the combined pyro- and hydrometallurgical process as described in Table 4 and Table 5

Source: JRC analysis

Table 7. Default recycling model for dismantling of PbA battery cells or monoblocs, inventory data per kg of battery cell

Component	Unit	Default value	Specification / dataset
Inputs			
EoL PbA battery	kg	1	--
Electricity	kWh	0.026	Electricity mix
Thermal energy from natural gas	MJ	0.09	Thermal energy from natural gas
Flocculants	kg	3.78×10^{-5}	Sodium silicate powder production
Limestone	kg	1.45×10^{-3}	Lime production; technology mix
Calcium hydroxide	kg	2.36×10^{-3}	Calcium hydroxide production; technology mix
Calcium oxide	kg	1.08×10^{-3}	Calcium hydroxide production, technology mix (proxy)
Soda (sodium carbonate)	kg	3.19×10^{-2}	Soda production; technology mix
Sodium hydroxide (100%; caustic soda)	kg	5.37×10^{-2}	Sodium hydroxide production; technology mix
Transport from the disassembling to the battery cell recycling plant	km	130	Transport by truck (>32 t, EURO 4)
Transport from the disassembling to the battery cell recycling plant	km	240	Transport by train (average freight train)
Transport from the disassembling to the battery cell recycling plant	km	270	Transport by ship (barge)
Outputs			
Lead containing material for refining	kg	0.82	To smelting and refining
Polyethylene (PE) [Waste for recovery]	kg	*	Incineration PE; waste-to-energy plant with dry flue gas treatment
Polypropylene (PP) [Waste for recovery]	kg	*	Incineration PP; waste-to-energy plant with dry flue gas treatment
Non-hazardous waste for incineration	kg	**	Waste incineration of inert material; waste-to-energy plant with dry flue gas treatment
Non-hazardous waste to landfill	kg	**	EF: Landfill of inert material (lead)
Water for wastewater treatment	kg	0.03	Treatment of residential wastewater, large plant
CO_2 emissions	kg	0.014	From oxidation of coke and chemical reactions (lime reduction)

* Plastic fraction to incineration according to its content in the battery cell and CFF parameters
** Other materials according to their content in the battery cell and CFF parameters

Source: JRC analysis.

Table 8. Default recycling model for smelting and refining of lead material from PbA battery cell dismantling, inventory data per kg of battery cell

Component	Unit	Default value	Specification / dataset
Inputs			
Lead containing material for refining	kg	0.82	From PbA cell dismantling
Electricity	kWh	0.105	Electricity mix
Thermal energy from natural gas	MJ	2.02	Thermal energy from natural gas
Thermal energy from heavy fuel oil	MJ	0.856	Thermal energy from heavy fuel oil
Oxygen gaseous	kg	0.115	
Petrol coke	kg	0.034	Coke; coke oven process, from hard coal
Calcium hydroxide	kg	0.032	Calcium hydroxide production; technology mix, 100% act. subst.
Iron oxide (II-oxide)	kg	8.93×10^{-3}	Iron oxide, red pigment; technology mix
Limestone	kg	1.65×10^{-3}	Lime production; technology mix
Sand	kg	7.82×10^{-3}	Silica sand, production mix
Soda (sodium carbonate)	kg	9.58×10^{-3}	Soda production, technology mix
Sodium hydroxide (100%; caustic soda)	kg	2.47×10^{-3}	Sodium hydroxide production; technology mix
Pyrite	kg	2.10×10^{-3}	
Outputs			
Lead (99.995%)	kg	0.633	
Hazardous waste to landfill (Pb dross)	kg	0.135	
Non-hazardous waste for land-filling	kg	0.131	
Non-hazardous waste for incineration	kg	0.013	
CO_2 emissions	kg	0.108	

Source: JRC own elaboration

Table 9. Default recycling model for recycling of Na-S battery, inventory data per kg of battery cell

Component	Unit	Default value	Dataset	Specification/ Dataset
Inputs				
EoL Na-S battery	kg	1	--	
Electricity	kWh	$1.6x10^{-3}$	Electricity mix	Battery preparation and dismantling
Nitrogen, gaseous	kg	0.026	Nitrogen, technology mix	Protective atmosphere
Water	kg	0.25	Water, deoinized	For oxidation of sodium metal and H_2SO_4 recovery
Outputs				
Recovered sodium (as NaOH)	kg	Credit according to content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat} and $Q_{Sout,r}/Q_p$ of the CFF.	sodium hydroxide, technology mix (Na recovered as NaOH)	
Recovered metals	kg	Credit for each metal according to its content in the battery cell and parameters R_{Return}, A_{Mat}, R_{rec,r_Mat} and $Q_{Sout,r}/Q_p$ of the CFF	Aluminium, steel	
Plastic components to incineration	kg		Incineration PP; waste-to-energy plant with dry flue gas treatment	typically, no plastic is contained in Na-S cells.
Other material to incineration	kg		Waste incineration of inert material; waste-to-energy plant with dry flue gas treatment	Other materials not recovered
Inert material to landfill	kg		Landfill of inert material (other materials)	Others fraction to landfill according to its content in the battery cell and CFF parameters
Hydrogen	kg		Emissions to air	From reaction of water with sodium

Source: JRC analysis. Inputs for shredding taken from (Blömeke *et al.*, 2022)

<u>Note</u>: The values in the default recycling processes are the result of intensive exchanges between JRC and stakeholders. The default recycling model for recycling of LIB, NiMH, SIB and other battery cells of similar layout (Table 4 and Table 5) correspond with those provided in the latest PEFCR for rechargeable batteries currently under development and public consultation.

6.3.2 The circular footprint formula ('CFF') applied to manufacturing waste

All manufacturing waste shall be considered in the modelling. The total amount of manufacturing waste shall be consistent with the bill of materials, the mass balance of the final product and the yield rates of the manufacturing processes.

The CFF shall be applied to all manufacturing waste using the same formula as in Section 6.3.1, with the following modifications:

— the equations for 'material input' and 'dismantling' shall not apply;

— the equation for 'electronics recycling' shall only apply if the manufacturing waste contains electronics/PWB waste;

— R_{Return} shall be equal to 1; and

— $E_{recEoL_Batterycell}$ shall only apply to waste fractions that consist of compound battery cell components, such as coated electrode cut-offs and final cells rejected at the end of the manufacturing line. In all other cases, $E_{recEoL_Batterycell}$ shall be equal to 0.

6.4 Transport

For transport in the main product production life cycle stage, in the distribution life cycle stage, and in the raw material acquisition and pre-processing life cycle stage between processes for which company-specific data is used pursuant to Sections 5.1, where relevant, and 5.2, company-specific data shall be used for the quantity of freight and the transport activity distance, which shall be determined in accordance with EN ISO standard 14083:2023 or equivalent, in combination with a secondary dataset corresponding to the actual transport mode and the actual category of the transport means, such the category of truck used.

For other transport in the raw material acquisition and pre-processing life cycle stage the manufacturer shall verify whether the datasets applied for that life cycle stage include all relevant transport. If that is not the case, the manufacturer shall complement these so that transport is accounted for, based on information from their own supply chain or based on average market data and supply chain analyses.

The categories of the transport means and the estimated distances shall be reported in the carbon footprint study.

7 Verification

7.1 Documentation

7.1.1 Public version of the CFB supporting study

The public version of the CFB study shall give meaningful information to consumers and other end-users on the calculation of the carbon footprint and shall include:

- information about the battery model;
- information about the geographical location of the battery manufacturing plant;
- the total life cycle carbon footprint expressed in kg CO_2-equivalent per kWh of total energy delivered (REP) or per kW·min·y of total backup service provided (OND);
- the carbon footprint of each life cycle stage listed in Section 4.2, expressed in kg CO_2-equivalent per kWh of total energy delivered (REP) or per kW·min·y of total backup service provided (OND);
- the reference year of the carbon footprint calculation i.e., the year for which the data for the CFB model was collected;
- the DQR score and the values of the single DQR criteria calculated in accordance with Section 5.5;
- the initial useable energy capacity, and the final useable energy capacity of the battery in kWh and its power in kW; for OND batteries the t_{se} in minutes;
- the total amount of CO_2-equivalent emitted during the life cycle of the battery, expressed in kg CO_2-equivalent per (unit of) battery, and the total amount of energy delivered by the battery over its service life as used for the carbon footprint calculation, in kWh (for REP) or the total amount of backup service provided by the battery over its service life as used for the carbon footprint calculation, in kW·min·y (for OND);
- the application the battery is designed for, the corresponding service life in terms of cycles and years, and the threshold for reaching the end of service life in percent of the battery's initial useable energy capacity;
- for REP-OtS batteries: the exact test procedure (including charge-discharge profiles over time, C-rates, temperature), the standard operating conditions and, if applicable, the underlying standard used for determining the battery lifetime;
- for OND-OtS batteries: the total backup service and the backup power capability, the lifetime in years, the standard or the exact test procedure (including charge-discharge profiles over time, C-rates, temperature) used for determining the battery service life (years of operation), and the standard operating conditions, and the threshold for determining the end of service life;
- for CM batteries: the average discharge power and discharge time in the application, and a list of all replacement parts needed for fulfilling the performance and lifetime as specified in the final ATS;
- information on each dataset used: the name of the dataset, for which life cycle stage or stages and which process or processes it was used, whether it is a company-specific dataset or a secondary dataset, its source in case of secondary datasets, its values of the DQR criteria, and its time validity;
- information on electricity modelling:

    - since for electricity modelling this report refers to the methodology specified in the final DA-EV, it is recommended to align with it also regarding the requirements for description and documentation of the electricity modelling;
- information on allocation:
    - indication of the multifunctional processes for which an allocation was applied;
    - which point of the hierarchy in Section 6.2.1 was applied in each case and a justification why any earlier points in the hierarchy could not be applied;
- information on recycled content and EoL modelling:
    - in case of recycled content higher than zero, information on the recycled content share;
    - in case of a company-specific return rate, information on the share of the batteries that are covered by an ownership business model;
    - the justification for modelling a company-specific battery cell recycling process, where applied.

Note: useable energy capacity is the energy available to the user when discharging a fully charged, new battery (at beginning of its service life) within the charge/ discharge limit set by the battery management system under standard operating conditions i.e., taking into account both charge limits and discharge limits.

7.1.2 Non-public version of the CFB supporting study

The carbon footprint study shall document in a systematic, orderly and comprehensive manner all steps taken for calculating the carbon footprint.

It shall include detailed description and documentation of:

- the carbon footprint model used for the calculation;
- any cut-off applied in the modelling, the resulting mass balance gap and an indication to which activity data or dataset the missing mass is assigned;
- the functional unit and all the information needed to calculate it, including the technical specification that the manufacturer provides to the customer (i.e., for CM batteries the complete final technical purchasing specification (TPS) and the corresponding final auditable technical specification (ATS) as referred to in the contractual agreement, and the corresponding contractual agreement; and for OtS the technical documentation delivered with the battery), containing, when applicable, the specified service life in years, the application load profile for which the battery is designed, including charge and discharge power, operation temperatures (at least average and 90% percentile), depth of discharge, maintenance requirements and a detailed list and description of all required replacements of system components, average number of cycles per day and per year, and the total energy that would be provided over lifetime of the system, including any test/simulation data and the associated commercial agreement.
- the details of all the company-specific data, including:
    - one or more tables containing all LCI, activity data and elementary flows, the corresponding background dataset used, if applicable, and the values of the DQR criteria of each dataset ('inventory tables'), both per functional unit and per kg of product. The tables shall include the list of parameters provided in Section 5.1, including the bill of materials or ingredients, such as substance names, units and quantities, information on origin, grades or purities, yield rates, where applicable,

recycled content and other technically or environmentally relevant characterisation, the procedure used for company-specific data collection, estimation and calculation, the data source where applicable, the data collection period, and a justification if data collection is done for periods of time other than the default timeframes specified in this document;

  - to which life cycle stage the data belong;
  - an indication of whether the data have been used for modelling the mandatory company-specific or non-mandatory company-specific processes;
  - all data sources, mathematical treatments applied to the data, and any underlying documentation needed to establish the reliability of the company-specific data.

— the details of all secondary and company-specific datasets used, including:
  - for non-mandatory most relevant processes, whether secondary or company-specific datasets were used, the method of selecting the datasets including any relevant assumptions and justifications such as the choice of a proxy in the case of TeR equalling four, which datasets were used, whether the electricity in the '-1 level' was changed and, in case it was changed, the justification for such change. In case company-specific datasets were used for non-mandatory most relevant processes in the raw material acquisition and pre-processing life cycle stage, the reference flow, the specific concentration of the target element (if applicable), and other substances contained as impurities in the main product (target material), in concentration or specific content of relevant elements, to allow a proper mass balance check;
  - for all secondary datasets the exact name of the dataset, the source, its universal unique identifier (UUID), location, dataset type ('LCI result' or 'partially terminated'), the values of the DQR criteria, and the time validity;
  - for all the company-specific datasets, the exact name of the dataset, its UUID, location, dataset type ('LCI result' or 'partially terminated'), the values of the DQR criteria, the time validity and the corresponding documentation for verification;

— the details of how the carbon footprint of electricity use was determined:
  - since for electricity modelling this report refers to the methodology as specified in the final DA-EV, it is recommended to align with it also regarding the requirements for description and documentation of the electricity modelling.

— the details of how allocation was applied, including:
  - indication of the multifunctional processes for which an allocation was applied and the allocation factors used;
  - which point of the hierarchy in Section 6.2.1 was applied in each case with a justification of why any earlier points in the hierarchy could not be applied;
  - in case of allocation based on physical relationships, which physical relationship has been used, a justification thereof, and the underlying data sources;
  - in case of economic allocation, the prices and the corresponding sources including the considered timespans used for the economic allocation, and justification for any timespans applied that are shorter than the default ones;
  - the datasets or activity data to which the allocation factors apply shall be indicated in the inventory tables;
  - if allocation is done for energy and auxiliary inputs of production lines as referred to in Section 6.2.2, the method applied and a justification thereof;

- the details of the recycled content and EoL modelling, including:
    - the parameters used in the CFF for all materials, with evidence for the cases where values other than the default values were used;
    - in case of recycled content higher than zero, the evidence for the recycled content share;
    - in case of a company-specific return rate, the evidence for the share of the batteries that are covered by an ownership business model where the property of the battery stays with the manufacturer together with evidence of the higher return rate, including evidence that the batteries covered under that rate will be recycled in accordance with Regulation (EU) 2023/1542;
    - in case of a company-specific battery cell recycling process, the evidence that the batteries will be recycled in a specific recycling plant and that the recycling process corresponds to the battery model for which the carbon footprint is declared;
- the categories of the transport means and the estimated distances;
- the public version of the carbon footprint study.

7.2 Verification and validation techniques

The notified body shall assess whether the carbon footprint declared meets the requirements set out in this section. It shall verify that:

a) data and information used for the calculation of the carbon footprint are consistent, reliable and traceable;

b) calculations performed are correct.

The assessment shall include a review of the carbon footprint study, the public version of the carbon footprint study, and, where appropriate, the model used to calculate the carbon footprint. For batteries manufactured in series, it shall include an assessment visit to:

a) the manufacturer's premises;

b) the cell, anode, and cathode production premises;

c) the cathode active material production premises;

d) the anode active material production premises; and

e) where considered important on the basis of the carbon footprint study, the premises of one or more of any other production sites for which company-specific data were collected.

The notified body shall verify that:

- all secondary datasets used in the model are appropriate and in compliance with the requirements laid out in this section;
- all company-specific data used in the model are appropriate and in compliance with the requirements laid out in this section, addressing in the verification:
    - coverage, precision, completeness, representativeness, consistency, reproducibility, sources and uncertainty;
    - plausibility, quality and accuracy of the data;
    - quality and accuracy of the underlying documentation.
- the electricity modelling rules as prescribed in Section 6.1 are correctly applied;

- calculations are of acceptable accuracy, reliable, are appropriate and performed in accordance with requirements laid out in this section;
- conversions of measurement units are correctly applied;
- methods for making estimates are appropriate and have been applied consistently;
- for CM batteries, that the battery fulfils the final TPS, and that the corresponding ATS fulfils the following criteria:
    - contains the customer requested load profile and relevant operating conditions, including the most stringent individual load profile;
    - describes the selected battery and the number of replacement battery modules or cells and any other replacement parts supplied in sequence along the specified service life;
    - indicates the final battery service life which shall be (under consideration of all replacement components) equal or greater than the requested application life specified in the TPS;
    - the ATS is part of the sales offer and the TPS is referred to in the contractual agreement.

The notified body shall identify uncertainties that are higher than expected and assess the effect of the identified uncertainty on the total carbon footprint.

8 Conclusions

The present report proposes a detailed framework for determining the carbon footprint of rechargeable industrial batteries with a capacity greater than 2 kWh, excluding those with exclusively external storage. As such, it is conceived as a technical basis for the further policy developments on the Delegated Act(s) that will ultimately define the official methodology to be followed for the CFB declaration under Article 7 of Regulation (EU) 2023/1542.

This report is the result of a two-year consensus-building process between the different stakeholders and aspires to represent best practice for calculating the carbon footprint of industrial batteries in line with objectives of Regulation (EU) 2023/1542. It considers the huge variety of battery technologies and applications contained in this category, while proposing a generic, mostly technology-agnostic approach. Also, it is aligned as far as possible with the draft Delegated Act defining the methodology for determining the carbon footprint of electric vehicle batteries, the ongoing work by the JRC on the definition of minimum durability requirements under Article 10 /Annex IV of Regulation (EU) 2023/1542 and international standardisation activities. This report is suggested to be used as a basis for developing the subsequent Delegated Act for industrial batteries under Article 7 of the Battery Regulation and also as guidance for future methodological developments and case studies related with life cycle assessment of batteries.

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List of abbreviations

A/C	Air Conditioning
AC	Alternating Current
AAM	Anode Active Material
ATS	Auditable Technical Specification
CAM	Cathode Active Material
CFB	Carbon Footprint of Batteries
CFB-EV	Carbon Footprint of Electric Vehicle Batteries
CFB-IND	Carbon Footprint of Industrial Batteries
CFF	Circular Footprint Formula
CM	Custom-Made (battery)
DA-EV	Delegated Act on carbon footprint of Electric Vehicle batteries
DQR	Data Quality Rating
DC	Direct Current
EF	Environmental Footprint
EoL	End of Life
EV	Electric Vehicle
FEqC	Full Equivalent Cycles
FSS	Fire Suppression System
HVAC	Heating, Ventilation and Air Conditioning
ILCD	International Reference Life Cycle Data system
IND	Industrial (battery)
JRC	Joint Research Centre
LCA	Life Cycle Assessment
LCDN	Life Cycle Data Network
LCI	Life Cycle Inventory
LIB	Lithium-Ion Battery
LMT	Light Means of Transport (batteries)
Na-S	Sodium-Sulphur (battery)
Na-NiCl	Sodium-Nickel Chloride (battery), also known as ZEBRA battery
NiCd	Nickel-Cadmium (battery)
NiMH	Nickel-Metal Hydride (battery)
NiZn	Nickel-Zinc (battery)
OND	On-demand (battery)
OtS	Off the Shelf (battery)
PbA	Lead Acid (battery)
PCS	Power Conversion System
PEF	Product Environmental Footprint
PEFCR	Product Environmental Footprint Category Rules

PWB	(populated) Printed Wiring Board
REP	Repetitive Energy Supply (battery)
SIB	Sodium-Ion Battery
SLI	Starter, Lighting, or Ignition (batteries)
TMS	Thermal Management System
TPS	Technical Purchasing Specification
UPS	Uninterruptible Power Supply

List of definitions

'Active material' refers to the materials within the battery cell which react chemically to produce electric energy when the battery cell discharges or to store electric energy when the battery is being charged. They are the main constituents of the anode and cathode, respectively. For lithium-ion batteries, anode active materials are typically graphites, sometimes doped with silicon, or spinels such as lithium titanate. Cathode active materials (CAM) are typically layered oxides such as Lithium Nickel Manganese Cobalt Oxide or Lithium Nickel Cobalt Aluminium Oxide, but also polyanionic materials such as Lithium Iron Phosphate.

'Active material precursor' means material required for the synthesis or production of active materials, either cathode active material precursors or anode active material precursors.

'Activity data' means the information associated with processes while modelling Life Cycle Inventories (LCI). The aggregated LCI results of the process chains that represent the activities of a process are each multiplied by the corresponding activity data and then combined to derive the carbon footprint associated with that process (Annex II Regulation (EU) 2023/1542).

'Auditable technical specification' (ATS), or auditable (battery) sizing report, means the technical specification that the battery manufacturer issues and that is provided together with the battery as delivered to the customer. It describes the exact characteristics of the delivered battery. The ATS is auditable i.e., made available to verifiers to ensure compliance. It explicitly refers to and fulfils or excels all requirements of the final TPS issued by the customer.

'Battery' means any device delivering electrical energy generated by direct conversion of chemical energy, having internal or external storage, and consisting of one or more non-rechargeable or rechargeable battery cells, modules or of packs of them (Article 3 of Regulation (EU) 2023/1542).

'Battery cell' means the basic functional unit in a battery constituted by electrodes, electrolyte, container, terminals and, if applicable, separators, and containing the active materials the reaction of which generates electrical energy (Article 3 of Regulation (EU) 2023/1542).

'Battery management system' means an electronic device that controls or manages the electric and thermal functions of a battery in order to ensure the battery's safety, performance and service life, manages and stores the data for the parameters for determining the battery's state of health and expected lifetime set out in Annex VII and communicates with the vehicle, light means of transport or appliance in which the battery is incorporated, or with a public or private charging infrastructure (Article 3 of Regulation (EU) 2023/1542).

'Battery manufacturing waste' means the materials or objects rejected during the battery manufacturing process, which cannot be re-used as an integral part in the same process and need to be recycled (Article 3 of Regulation (EU) 2023/1542).

'Battery module' means any set of battery cells that are connected together or encapsulated within an outer casing to protect the cells against external impact, and which is meant to be used either alone or in combination with other modules (Article 3 of Regulation (EU) 2023/1542).

'Battery pack' means any set of battery cells or modules that are connected together or encapsulated within an outer casing, to form a complete unit which is not meant to be split up or opened by the end-user (Article 3 of Regulation (EU) 2023/1542).

'Battery model' means a version of a battery all units of which share the same technical characteristics relevant for the requirements of this Regulation on sustainability, safety, labelling, marking and information, and the same model identifier (Article 3 of Regulation (EU) 2023/1542).

'Beginning of service life' means the moment when a new battery starts its operation i.e., just before it provides its first charge-discharge cycle within the application.

'Bill of materials' means a list of the raw materials, sub-assemblies, intermediate assemblies, sub-components and parts, and the quantities of each, needed to manufacture the battery (Regulation (EU) 2023/1542 - Annex II).

'Carbon footprint' means the sum of greenhouse gas emissions and greenhouse gas removals in a product system, expressed as carbon dioxide equivalents and based on a Product Environmental Footprint

(PEF) study using the single impact category of climate change (Article 3 of Regulation (EU) 2023/1542).

'CFB-EV' indicates, in this report, the value of the carbon footprint to be declared for electric vehicle batteries, according to Article 7 of the Regulation (EU) 2023/1542.

'CFB-IND' indicates, in this report, the value of the carbon footprint to be declared for rechargeable industrial batteries with a capacity greater than 2 kWh except those with exclusively external storage, according to Article 7 of the Regulation (EU) 2023/1542.

'CFB-IND rules' indicates, in this report, the methodological guidelines developed in the present document and needed to calculate the value of the CFB-IND.

'CFB-compliant company-specific dataset' means a company-specific dataset that respects the requirements defined in Section 5.3 of this document. A CFB-compliant company-specific dataset can be either a LCI dataset or a LCIA dataset.

'Characterisation factor' means a factor derived from a characterisation model which is applied to convert an assigned life cycle inventory result to the common unit of a LCIA impact category.

'Circular Footprint Formula' describes how burdens and benefits from disposal and recovery of the product or service assessed, as well as use of secondary materials (i.e., recycled content) into that product or service, are allocated to the system under study.

'Climate change' is the LCIA impact category considering all inputs and outputs that result in greenhouse gas (GHG) emissions. The consequences include increased average global temperatures and sudden regional climatic changes.

'Company-specific dataset' means a dataset (disaggregated or aggregated) compiled with company-specific data. The activity data and direct elementary exchanges of company-specific data sets shall refer to company-specific information, while the underlying sub-processes may be modelled with company-specific or secondary datasets.

'Dataset' means a document or file with life cycle information of a specified product or other reference such as a process, covering its descriptive metadata and either its quantitative life cycle inventory in case of a life cycle inventory ('LCI') result dataset or its carbon footprint in case of a life cycle impact assessment ('LCIA') result dataset.

'Data collection period' means the date(s) or time period(s) when the data was collected. Note that this does NOT refer to e.g., the publication dates of papers or books from which the data may stem, but to the original data collection period.

'EF-compliant dataset' means dataset developed in compliance with the EF requirements in terms of modelling and methodological compliance (in agreement with the "Commission Recommendation (EU) 2021/2279"); meta data compliance (in agreement with Fazio et al. 2020, "Guide on EF compliant datasets"); nomenclature, and characterization factors in agreement with the "EF 3.1 Reference Package"). The DQR of each single data quality indicator shall be lower or equal than 3.

'Elementary flows' means the materials or energy entering the system being studied that have been drawn from the environment without previous human transformation, or material or energy leaving the system being studied that is released into the environment without subsequent human transformation.

'End of service life' means the state of the battery where it is, due to ageing or other degradation processes, considered unsuitable for continuing to provide service within the primary application and where it is decommissioned or discarded and either sent for recycling or for other treatment (such as refurbishing, remanufacturing, preparation for re-use or preparation for repurposing) for subsequent use in a second life application.

'Final useable energy capacity' means the energy available to the user when discharging a fully charged battery at its end of service life within the charge/ discharge limit set by the battery management system under standard operating conditions (such as maximum depth of discharge for increasing lifetime). Fully charged refers to charging until the charge level set by the battery management system under standard operating conditions.

‘Full cycles’ is the number of times the battery can fully discharge within the charge/discharge limits set by the battery management system under standard operating conditions. The actual energy delivered in each full cycle decreases over lifetime. For a new battery (at beginning of its service life), one full cycle is equivalent to one full equivalent cycle (FEqC). The full cycles multiplied with the initial useable energy capacity and the average value between 100% and the state of health (or remaining useable energy capacity in %) at EoL gives the total energy provided by the battery over lifetime.

‘Full equivalent cycles’ (FEqC) is the number of times the battery would theoretically be able to discharge its initial useable energy capacity. The FEqC multiplied with the initial useable energy capacity gives the total energy provided by the battery over lifetime.

‘Functional unit’ means the qualitative and quantitative aspects of the functions, services, or both, provided by the battery (Annex II of Regulation (EU) 2023/1542).

‘ILCD entry-level compliant dataset’ means a dataset developed in accordance with the compliance rules and entry-level requirements of the International Reference Lifecycle Data System (‘ILCD’) Data Network, available on the European Platform on LCA, and in accordance with the EF Reference Package, available in the Life Cycle Data Network (‘LCDN’) on the European Platform of LCA, for nomenclature and characterisation factors compliance.

‘Initial useable energy capacity’ means the energy available to the user when discharging a fully charged, new battery (i.e., at the beginning of its service life) within the charge/ discharge limits set by the battery management system under standard operating conditions (such as maximum depth of discharge for increasing lifetime). “Fully charged” refers to charging until the charge level set by the battery management system under standard operating conditions.

‘Intermediate product’ means the output form of a unit process that in turn is input to other unit processes which require further transformation within the system. An intermediate product is a product that requires further processing before it is saleable to the final consumer.

‘LCI result dataset’ means a dataset modelling a complete or partial life cycle of a product system that, apart from the product flow associated with the reference product, lists only elementary flows.

‘LCIA result dataset’ means a dataset modelling a complete or partial life cycle of a product system that, apart from the product flow associated with the reference product, lists only the result of the life cycle impact assessment, which in the case where climate change is the only impact category analysed means that the result is the amount of carbon dioxide (‘CO_2’)-equivalent.

‘Life cycle’ means the consecutive and interlinked stages of a product system, from raw material acquisition or generation from natural resources to final disposal (ISO 14040:2006 or equivalent) (Annex II of Regulation (EU) 2023/1542).

‘Life cycle inventory (LCI)’ means the combined set of exchanges of elementary, waste and product flows in a LCI dataset (Annex II of Regulation (EU) 2023/1542).

‘Life cycle inventory (LCI) dataset’ means a document or file with life cycle information on a specified product or other reference, such as the site or process, covering descriptive metadata and quantitative life cycle inventory, which could include a unit process dataset, partially disaggregated or an aggregated dataset.

‘Lifetime of a battery’ means the period that starts when the battery is manufactured and ends when the battery becomes waste (Article 3 of Regulation (EU) 2023/1542).

‘Manufacturer’ means any natural or legal person who manufactures a battery or has a battery designed or manufactured, and markets that battery under its own name or trademark or puts it into service for its own purposes (Article 3 of Regulation (EU) 2023/1542).

‘Most representative secondary dataset’ means the secondary dataset having the highest technological representativeness (‘TeR’) quality rating or, if there are several datasets with the same TeR, the one with the highest geographical representativeness (‘GeR’) quality rating, or, in the absence of information on the geographical provenience of the material concerned, the one representing the global average.

‘Notified body’ means a conformity assessment body that has been notified in accordance with Chapter V of Regulation (EU) 2023/1542.

Partial cycles is the number of times the battery can discharge with a given discharge profile, but without fully discharging within the charge/discharge limits set by the battery management system.

'Partially disaggregated dataset' means a dataset with a LCI that contains elementary flows and activity data, and that only in combination with its complementing underlying datasets yields a complete aggregated LCI dataset.

'Partially disaggregated dataset at -1 level' means a partially disaggregated dataset that contains elementary flows and activity data for one level down in the supply-chain, while all complementing underlying datasets are in their aggregated form.

'Placing on the market' means the first making available of a battery on the Union market (Article 3 of Regulation (EU) 2023/1542).

'Recycler' means any natural or legal person who carries out recycling in a permitted facility (Article 3 of Regulation (EU) 2023/1542).

'Reference flow' means the measure of the outputs from processes in a given product system required to fulfil the function expressed by the functional unit.

'Secondary data' means data not from a specific process within the supply-chain of the company performing a carbon footprint study. This refers to data that is not directly collected, measured, or estimated by the company, but sourced from a third party LCI database or other sources. Secondary data includes industry average data (e.g., from published production data, government statistics, and industry associations), literature studies, engineering studies and patents, and may also be based on financial data, and contain proxy data, and other generic data.

'Secondary dataset' refers to any dataset that is not company-specific, e.g., from a database.

'State of health' means a measure of the general condition of a rechargeable battery and its ability to deliver the specified performance compared with its initial condition (Article 3 of Regulation (EU) 2023/1542).

'System boundary' indicates which aspects are included or excluded from the life cycle study. The system boundary defines which parts of the product life cycle and which associated life cycle stages and processes belong to the analysed system (i.e., are required for carrying out its function as defined by the functional unit), except for those processes excluded based on the cut-off rule.

'Technical purchasing specification' (TPS) means the technical specification that is issued by the customer and that contains the application requirements for a CM battery. The final TPS (in case the initial TPS is modified as a result of negotiations or the received offers with suppliers) describes the service and performance the CM battery has to deliver to the application.

'Technical specification' means a document that prescribes technical requirements to be fulfilled by a product, process or service (Article 3 of Regulation (EU) 2023/1542) and that is delivered together with the battery to the customer or end-user.

'Treatment' means any operation carried out on waste batteries after they have been handed over to a facility for sorting, preparation for re-use, preparation for repurposing, preparation for recycling or for recycling (Article 3 of Regulation (EU) 2023/1542).

'Unit process' means the smallest element considered in the LCI for which input and output data are quantified (based on ISO 14040:2006).

Apart from the previously listed, the definitions provided in Regulation (EU) 2023/1542 apply.

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